

# MODELLING THE HOUSING MARKET IN IRELAND



**Author:** Laurel Dsouza

**Supervisor:** Dr Kieran Mulchrone

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School of Mathematical Sciences,  
University College Cork,  
Ireland

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# Declaration of Authorship

This report is wholly the work of the author, except where explicitly stated otherwise. The source of any material which was not created by the author has been clearly cited.

**Date:** 02/10/2024

**Signature:** Laurel Dsouza

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# Abstract

The Irish housing market faces unprecedented challenges, with population growth outpacing new home construction by nearly four to one, the worst ratio among nine high-income countries studied [1]. This thesis presents an agent-based model (ABM) of the Irish housing market, designed to capture the complex interactions between various factors influencing housing dynamics.

Using data from the Central Statistics Office of Ireland, which reports an 8.6% year-on-year increase in residential property prices as of 2024 [1], the ABM simulates the behavior of individual agents (households, banks, and construction companies) to model emergent market behavior. The research aims to provide a more nuanced understanding of housing market dynamics than traditional economic models, offering a platform for testing various policy interventions.

The model incorporates key elements such as population changes, income profiles, housing stock, new housing development, bank lending policies, and interest rates. It explores scenarios ranging from the current government target of 33,000 new homes annually to the Economic and Social Research Institute's projection of up to 53,000 homes needed yearly [1].

This research contributes to both the academic understanding of housing markets and practical policy-making, offering a tool for predicting market trends and evaluating the potential impacts of policy interventions in the Irish context. The findings aim to inform strategies for addressing Ireland's severe affordability decline and supply shortage in the housing sector.

# Contents

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                        | <b>11</b> |
| 1.1      | Background . . . . .                                       | 11        |
| 1.2      | Problem Statement . . . . .                                | 11        |
| 1.3      | Research Objectives . . . . .                              | 12        |
| 1.4      | Methodology Overview . . . . .                             | 12        |
| 1.5      | Significance of the Study . . . . .                        | 12        |
| 1.6      | Thesis Structure . . . . .                                 | 13        |
| <br>     |                                                            |           |
| <b>2</b> | <b>Theoretical Background</b>                              | <b>14</b> |
| 2.1      | Agent-Based Modeling . . . . .                             | 14        |
| 2.1.1    | Definition and Key Concepts . . . . .                      | 14        |
| 2.1.2    | Advantages and Limitations of ABM . . . . .                | 15        |
| 2.1.3    | Applications in Economics and Social Sciences . . . . .    | 16        |
| 2.1.4    | ABM and Complexity Theory . . . . .                        | 16        |
| 2.1.5    | Developing Agent-Based Models . . . . .                    | 17        |
| 2.1.6    | Software for Agent-Based Modeling . . . . .                | 17        |
| 2.2      | Housing Market Dynamics . . . . .                          | 17        |
| 2.2.1    | Supply and Demand in Housing Markets . . . . .             | 17        |
| 2.2.2    | Factors Influencing Housing Prices . . . . .               | 18        |
| 2.2.3    | Role of Banks and Lending Policies . . . . .               | 18        |
| 2.2.4    | Impact of Government Policies on Housing Markets . . . . . | 18        |
| 2.3      | Irish Housing Market Context . . . . .                     | 19        |
| 2.3.1    | Historical Overview of Irish Housing Market . . . . .      | 19        |
| 2.3.2    | Current Challenges in Irish Housing . . . . .              | 19        |

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| 2.3.3    | Key Stakeholders in the Irish Housing Market . . . . .     | 19        |
| 2.3.4    | Relevant Irish Housing Policies and Regulations . . . . .  | 20        |
| 2.4      | Integration of ABM in Housing Market Analysis . . . . .    | 20        |
| 2.4.1    | Previous ABM Studies on Housing Markets . . . . .          | 20        |
| 2.4.2    | Key Components of Housing Market ABMs . . . . .            | 21        |
| 2.4.3    | Modeling Housing Market Dynamics . . . . .                 | 21        |
| 2.4.4    | ABM Potential for Irish Housing Analysis . . . . .         | 22        |
| 2.4.5    | Challenges and Considerations . . . . .                    | 23        |
| <b>3</b> | <b>Literature Review</b>                                   | <b>24</b> |
| 3.1      | Introduction . . . . .                                     | 24        |
| 3.2      | Foundations of Agent-Based Modeling in Housing Markets . . | 25        |
| 3.2.1    | Pioneering Studies . . . . .                               | 25        |
| 3.3      | Comprehensive Market Models . . . . .                      | 25        |
| 3.3.1    | UK Housing Market Model . . . . .                          | 26        |
| 3.3.2    | Dutch Housing Market Model . . . . .                       | 26        |
| 3.3.3    | High-Resolution Data-Driven Model . . . . .                | 27        |
| 3.3.4    | Synthesis and Contribution . . . . .                       | 27        |
| 3.4      | Modeling Human Behavior in Housing Markets . . . . .       | 28        |
| 3.4.1    | Utility-Based Decision Making . . . . .                    | 28        |
| 3.4.2    | Behavioral Economics in ABMs . . . . .                     | 28        |
| 3.5      | Spatial Considerations in Housing Market ABMs . . . . .    | 29        |
| 3.5.1    | Importance of Location . . . . .                           | 29        |
| 3.5.2    | Challenges in Spatial Modeling . . . . .                   | 29        |
| 3.6      | Construction and Market Dynamics . . . . .                 | 29        |
| 3.6.1    | Modeling Construction . . . . .                            | 29        |
| 3.6.2    | Supply-Demand Mismatch . . . . .                           | 30        |
| 3.7      | Social Housing and Policy Implications . . . . .           | 30        |
| 3.7.1    | Importance of Social Housing . . . . .                     | 30        |
| 3.7.2    | Policy Modeling . . . . .                                  | 30        |
| 3.8      | Model Design, Validation, and Challenges . . . . .         | 30        |
| 3.8.1    | Design Considerations . . . . .                            | 31        |
| 3.8.2    | Validation Approaches . . . . .                            | 31        |

|          |                                            |           |
|----------|--------------------------------------------|-----------|
| 3.8.3    | Computational Challenges . . . . .         | 31        |
| <b>4</b> | <b>Model Development and Methodology</b>   | <b>32</b> |
| 4.1      | Model Overview . . . . .                   | 32        |
| 4.2      | Agents and Their Characteristics . . . . . | 33        |
| 4.2.1    | Households . . . . .                       | 33        |
| 4.2.1.1  | Properties . . . . .                       | 33        |
| 4.2.1.2  | Initialization . . . . .                   | 34        |
| 4.2.1.3  | Decision-making processes . . . . .        | 35        |
| 4.2.1.4  | Wealth and Income Updates . . . . .        | 38        |
| 4.2.1.5  | Limitations and Assumptions . . . . .      | 39        |
| 4.2.2    | Houses . . . . .                           | 40        |
| 4.2.2.1  | Properties . . . . .                       | 40        |
| 4.2.2.2  | Initialization and Construction . . . . .  | 41        |
| 4.2.2.3  | Key Methods and Processes . . . . .        | 42        |
| 4.2.2.4  | Limitations and Assumptions . . . . .      | 44        |
| 4.2.3    | Bank . . . . .                             | 45        |
| 4.2.3.1  | Properties . . . . .                       | 45        |
| 4.2.3.2  | Initialization . . . . .                   | 45        |
| 4.2.3.3  | Key Methods and Processes . . . . .        | 45        |
| 4.2.3.4  | Limitations and Assumptions . . . . .      | 47        |
| 4.2.4    | Construction Company . . . . .             | 47        |
| 4.2.4.1  | Properties . . . . .                       | 47        |
| 4.2.4.2  | Initialization . . . . .                   | 48        |
| 4.2.4.3  | Key Methods and Processes . . . . .        | 48        |
| 4.2.4.4  | Market Dynamics . . . . .                  | 49        |
| 4.2.4.5  | Limitations and Assumptions . . . . .      | 50        |
| 4.3      | Simulation Process . . . . .               | 51        |
| 4.3.1    | Simulation Overview . . . . .              | 51        |
| 4.3.2    | Initialization . . . . .                   | 52        |
| 4.3.2.1  | Initialize(scaleFactor) . . . . .          | 52        |
| 4.3.3    | Main Simulation Loop . . . . .             | 52        |
| 4.3.4    | Monthly Processes . . . . .                | 53        |

|         |                                            |    |
|---------|--------------------------------------------|----|
| 4.3.4.1 | Update House Prices . . . . .              | 53 |
| 4.3.4.2 | Update Contracts . . . . .                 | 53 |
| 4.3.4.3 | Decide To Move . . . . .                   | 53 |
| 4.3.4.4 | Act On Best Options . . . . .              | 54 |
| 4.3.4.5 | Assign Social Housing . . . . .            | 54 |
| 4.3.4.6 | Process New Listings . . . . .             | 54 |
| 4.3.4.7 | Collect Payments . . . . .                 | 54 |
| 4.3.5   | Yearly Processes . . . . .                 | 54 |
| 4.3.5.1 | Update Population . . . . .                | 55 |
| 4.3.5.2 | Update Incomes . . . . .                   | 55 |
| 4.3.5.3 | Set Tax Rate . . . . .                     | 55 |
| 4.3.5.4 | Construct New Houses . . . . .             | 55 |
| 4.3.6   | Data Collection and Analysis . . . . .     | 56 |
| 4.3.6.1 | Output Monthly Statistics . . . . .        | 56 |
| 4.3.6.2 | Output Yearly Statistics . . . . .         | 56 |
| 4.3.6.3 | Calculate Statistics . . . . .             | 56 |
| 4.4     | Model Parameters and Assumptions . . . . . | 56 |
| 4.4.1   | Key Parameters . . . . .                   | 57 |
| 4.4.1.1 | Temporal Parameters . . . . .              | 57 |
| 4.4.1.2 | Household Parameters . . . . .             | 57 |
| 4.4.1.3 | Housing Parameters . . . . .               | 57 |
| 4.4.1.4 | Economic Parameters . . . . .              | 58 |
| 4.4.1.5 | Market Dynamics Parameters . . . . .       | 58 |
| 4.4.2   | Critical Assumptions . . . . .             | 58 |
| 4.4.2.1 | Household Behavior . . . . .               | 58 |
| 4.4.2.2 | Housing Market . . . . .                   | 58 |
| 4.4.2.3 | Economic Factors . . . . .                 | 59 |
| 4.4.2.4 | Government Policies . . . . .              | 59 |
| 4.4.3   | Limitations . . . . .                      | 59 |
| 4.5     | Data and Data Sources . . . . .            | 60 |
| 4.5.1   | Demographic Data . . . . .                 | 60 |
| 4.5.1.1 | Household Types . . . . .                  | 60 |
| 4.5.1.2 | Children Distribution . . . . .            | 60 |

|          |                                                             |           |
|----------|-------------------------------------------------------------|-----------|
| 4.5.1.3  | Child Age Groups . . . . .                                  | 61        |
| 4.5.1.4  | Adult Age Groups . . . . .                                  | 61        |
| 4.5.1.5  | Housing Types . . . . .                                     | 61        |
| 4.5.2    | Income Data . . . . .                                       | 62        |
| 4.5.2.1  | Mean Income by Age Group and Gender (2022)                  | 62        |
| 4.5.3    | Data Notes . . . . .                                        | 62        |
| <b>5</b> | <b>Results</b>                                              | <b>63</b> |
| 5.1      | Model Validation . . . . .                                  | 63        |
| 5.2      | Overview of Simulation Outcomes . . . . .                   | 65        |
| 5.3      | Key Findings . . . . .                                      | 69        |
| 5.3.1    | Market Activity . . . . .                                   | 69        |
| 5.3.2    | Market Stability . . . . .                                  | 70        |
| 5.3.3    | Long-term Effects . . . . .                                 | 71        |
| 5.3.4    | Homeownership Rates . . . . .                               | 71        |
| 5.4      | Interpretation . . . . .                                    | 72        |
| 5.5      | Impact of Limited Construction and Restricted Vacancy . . . | 74        |
| 5.5.1    | Key Observations . . . . .                                  | 74        |
| 5.5.2    | Interpretation . . . . .                                    | 75        |
| 5.5.3    | Comparison with Unconstrained Scenarios . . . . .           | 76        |
| 5.6      | Comparison Across Scenarios . . . . .                       | 77        |
| 5.7      | Conclusion . . . . .                                        | 77        |
| 5.8      | Discussion . . . . .                                        | 80        |
| 5.8.1    | Theoretical Implications . . . . .                          | 80        |
| 5.8.2    | Methodological Reflections . . . . .                        | 81        |
| 5.8.3    | Ethical Considerations . . . . .                            | 81        |
| 5.8.4    | Future Research Directions . . . . .                        | 82        |
| <b>A</b> | <b>Code</b>                                                 | <b>83</b> |
| A.0.1    | Simulation Class . . . . .                                  | 83        |
| A.0.1.1  | Run Method . . . . .                                        | 83        |
| A.0.2    | SimulationUpdater Class . . . . .                           | 84        |
| A.0.2.1  | PerformMonthlyUpdates Method . . . . .                      | 84        |

|         |                                     |    |
|---------|-------------------------------------|----|
| A.0.2.2 | DecideToMove Method . . . . .       | 85 |
| A.0.3   | Household Class . . . . .           | 86 |
| A.0.3.1 | DecideToMove Method . . . . .       | 86 |
| A.0.4   | House Class . . . . .               | 87 |
| A.0.4.1 | UpdatePrice Method . . . . .        | 87 |
| A.0.5   | Bank Class . . . . .                | 88 |
| A.0.5.1 | ApproveMortgage Method . . . . .    | 88 |
| A.0.6   | ConstructionCompany Class . . . . . | 88 |
| A.0.6.1 | ConstructNewHouses Method . . . . . | 88 |

# List of Figures

|     |                                                                                                  |    |
|-----|--------------------------------------------------------------------------------------------------|----|
| 5.1 | Original Data . . . . .                                                                          | 64 |
| 5.2 | Comparison of Contract Types: Original vs. Simulation-Generated<br>Data . . . . .                | 64 |
| 5.3 | Households Wanting to Move Over Time: 3% Interest Rate . .                                       | 66 |
| 5.4 | Households Wanting to Move Over Time: 6% Interest Rate . .                                       | 66 |
| 5.5 | Homeownership Rate Over Time: 3% Interest Rate . . . . .                                         | 67 |
| 5.6 | Homeownership Rate Over Time: 6% Interest Rate . . . . .                                         | 67 |
| 5.7 | Households Wanting to Move Over Time: Limited Construc-<br>tion and Restricted Vacancy . . . . . | 68 |
| 5.8 | Homeownership Rate Over Time: Limited Construction and<br>Restricted Vacancy . . . . .           | 68 |
| 5.9 | Households Wanting to Move Over Time: Limited Construc-<br>tion and Restricted Vacancy . . . . . | 74 |

# Chapter 1

## Introduction

### 1.1 Background

Housing is a fundamental necessity and a critical component of any functioning society. In recent years, housing has become a pressing issue in Ireland, dominating political discourse and media coverage. The complexity of the housing market, influenced by numerous interconnected factors such as population dynamics, economic conditions, and policy decisions, makes it an ideal candidate for analysis through mathematical modelling and computational simulation techniques.

### 1.2 Problem Statement

The Irish housing market faces multiple challenges, including affordability issues, supply shortages, and the need for effective policy interventions. Traditional economic models, often struggle to capture the full complexity of housing markets, particularly the heterogeneous nature of households and properties, and the dynamic interactions between various stakeholders. This research aims to address this gap by developing an agent-based model (ABM) of the Irish housing market, capable of simulating the complex interactions and emergent behaviors that characterize real-world housing dynamics.

## 1.3 Research Objectives

The main objectives of this study are:

- To develop a comprehensive agent-based model of the Irish housing market, incorporating key factors such as population changes, income profiles, housing stock, new housing development, bank lending policies, and interest rates.
- To calibrate the model using real-world data from Ireland, ensuring its relevance and accuracy in representing the current housing situation.
- To explore potential policy interventions and their impacts on the housing market, providing data-driven insights to guide policy-making.

## 1.4 Methodology Overview

This research employs agent-based modelling, a computational method that simulates the actions and interactions of autonomous agents within a system. The model is developed using C# programming language, incorporating data from the Central Statistics Office of Ireland and other relevant sources. The simulation includes various types of agents (e.g., households, banks, construction companies) and models their behaviors and interactions over time.

## 1.5 Significance of the Study

This research contributes to both the academic understanding of housing markets and practical policy-making. By providing a detailed, bottom-up simulation of the Irish housing market, it offers insights into the complex dynamics that traditional models may overlook. The ability to test various scenarios and policy interventions in a simulated environment can inform more effective and targeted housing policies, potentially leading to improved outcomes for Irish citizens in terms of housing affordability and availability.

## 1.6 Thesis Structure

This thesis is organized as follows:

- Chapter 2 provides the theoretical background on agent-based modeling and housing market dynamics, including foundations of ABM in housing markets, spatial considerations, and construction dynamics.
- Chapter 3 reviews relevant literature on agent-based models of housing markets, including pioneering studies, comprehensive market models, and approaches to modeling human behavior in housing markets.
- Chapter 4 presents the detailed methodology, including model overview, agent characteristics, simulation process, model parameters and assumptions, and data sources used in the model.
- Chapter 5 showcases the results of the simulation, including model validation and outcomes from three scenarios.
- Chapter 6 discusses the findings, their theoretical implications, methodological reflections, ethical considerations, and potential future research directions.

# Chapter 2

## Theoretical Background

This section provides a comprehensive theoretical background for the study, covering three main areas. It begins with an in-depth exploration of Agent-Based Modeling (ABM), detailing its definition, key concepts, advantages, limitations, and applications in economics and social sciences. The section then delves into housing market dynamics, examining factors that influence supply, demand, and prices, including the roles of banks and government policies. Finally, it contextualizes the Irish housing market, offering a historical overview, outlining current challenges, identifying key stakeholders, and discussing relevant policies and regulations.

### 2.1 Agent-Based Modeling

#### 2.1.1 Definition and Key Concepts

Agent-Based Modeling (ABM) is a computational approach for simulating the actions and interactions of autonomous agents within a system. In this study, agents represent entities such as households, banks, and construction companies. ABM enables the modeling of complex systems from the bottom up, capturing emergent phenomena arising from individual agents' interactions [2].

Key concepts in ABM include:

- **Agents:** Autonomous entities with defined attributes and behaviors. Agents typically possess four critical characteristics [3]:
  - **Perception:** The ability to perceive their environment and other agents.
  - **Performance:** A set of behaviors they can execute.
  - **Memory:** The capacity to record previous states and actions.
  - **Policy:** A set of rules or strategies that guide their next actions.
- **Environment:** The space in which agents interact, which can be either abstract or represent real-world geographical spaces.
- **Rules:** The behaviors governing agent actions and interactions.
- **Emergence:** The development of complex patterns and phenomena from simpler individual interactions.

### 2.1.2 Advantages and Limitations of ABM

ABM provides several advantages for studying complex systems:

- It allows for capturing heterogeneity among agents, which can lead to more realistic models [4].
- It offers flexibility in modeling non-linear behaviors and interactions [2].
- ABM has the capacity to generate emergent phenomena, reflecting complex dynamics from simple rules [5].
- It makes it easier to implement spatial and network effects, allowing for geographically and socially embedded models [6].

However, ABM also presents some limitations:

- It can be computationally intensive, especially when dealing with large-scale models [4].

- ABM models face challenges in validation and calibration, as they often rely on numerous parameters [7].
- Models can be sensitive to initial conditions and parameter settings, potentially affecting the robustness of the results [8].

### 2.1.3 Applications in Economics and Social Sciences

ABM has been widely employed across various domains in economics and social sciences, including:

- Financial markets, to model market dynamics and investor behavior [9].
- Consumer behavior, to simulate purchasing patterns and decision-making processes [10].
- Urban planning, particularly in simulating the growth and development of cities [11].
- Housing markets, capturing the interaction between demand, supply, and price dynamics [9].

These applications illustrate the versatility of ABM in modeling complex social and economic systems, making it an appropriate choice for analyzing the Irish housing market.

### 2.1.4 ABM and Complexity Theory

Agent-Based Modeling (ABM) is closely linked to complexity theory, reflecting the increasing interest in complex adaptive systems among social scientists. Complex systems are defined by their high degree of complexity, path dependence, non-linearity, and self-organization [12]. ABM provides a powerful framework to study these systems, enabling researchers to simulate how macroscopic regularities emerge over time from the interactions of autonomous and heterogeneous agents [9].

### 2.1.5 Developing Agent-Based Models

The process of developing an agent-based model typically involves several key steps [11]:

1. **Specification and Formalization:** Defining the research question, reviewing relevant theories, and formalizing the model's design.
2. **Modeling, Verification, and Experimentation:** Implementing the model, debugging, and conducting experiments to analyze system behavior.
3. **Calibration and Validation:** Comparing the model's outputs with real-world data and refining the model to improve its accuracy and predictive power.

### 2.1.6 Software for Agent-Based Modeling

Several software tools are available to assist in the development of agent-based models. Some of the most widely used platforms include:

- **Swarm**
- **RePast**
- **Mason**
- **NetLogo**

Among these, **NetLogo** is particularly recommended for beginners due to its user-friendly interface, extensive documentation, and active user community [13].

## 2.2 Housing Market Dynamics

### 2.2.1 Supply and Demand in Housing Markets

The dynamics of housing markets are fundamentally shaped by the interaction of supply and demand. On the supply side, factors such as the availabil-

ity of land, construction costs, and regulatory frameworks play pivotal roles. Conversely, demand is primarily driven by variables like population growth, income levels, and changing consumer preferences [14].

### 2.2.2 Factors Influencing Housing Prices

Housing prices are influenced by a complex array of economic, social, and policy factors. Key drivers include:

- **Economic conditions:** Factors such as GDP growth and employment rates.
- **Demographic trends:** Changes in population size and composition.
- **Interest rates and mortgage availability:** The cost and accessibility of borrowing can significantly impact demand.
- **Government policies and regulations:** Zoning laws, tax incentives, and housing subsidies.
- **Location and amenities:** Proximity to employment centers, schools, and other amenities increases property values.

### 2.2.3 Role of Banks and Lending Policies

Banks and financial institutions play a crucial role in housing markets through their mortgage lending policies. The availability of mortgage financing, interest rates, and loan terms directly influence housing demand and prices. Changes in lending practices can lead to fluctuations in market activity, influencing both short-term demand and long-term price trends [15].

### 2.2.4 Impact of Government Policies on Housing Markets

Government interventions, including zoning laws, tax policies, and housing subsidies, significantly affect housing market dynamics. Policies that regulate

land use or incentivize certain housing developments can affect both supply and demand, leading to shifts in housing prices and availability [16].

## **2.3 Irish Housing Market Context**

### **2.3.1 Historical Overview of Irish Housing Market**

The Irish housing market has undergone dramatic fluctuations over the past few decades, notably during the boom and bust cycle of the Celtic Tiger era. The rapid expansion of housing construction in the early 2000s was followed by a significant collapse in 2008, which resulted in a prolonged period of housing market instability [17].

### **2.3.2 Current Challenges in Irish Housing**

The Irish housing market faces several pressing challenges, including:

- A severe housing shortage, particularly in urban areas such as Dublin.
- Affordability issues, especially for first-time buyers who are struggling to enter the market.
- A rapid increase in rental prices, driven by high demand and limited supply.
- Rising homelessness and housing insecurity, exacerbated by inadequate social housing provision.

### **2.3.3 Key Stakeholders in the Irish Housing Market**

The main stakeholders influencing the Irish housing market include:

- **Households:** Individuals and families seeking to buy or rent housing.
- **Property Developers:** Firms involved in building and selling residential properties.

- **Financial Institutions:** Banks providing mortgage financing to prospective buyers.
- **Government Agencies:** Bodies responsible for housing policy, regulation and social housing provision.
- **Non-Profit Organizations:** Groups working to provide affordable housing and address homelessness [18].

### 2.3.4 Relevant Irish Housing Policies and Regulations

Recent government initiatives aimed at addressing the housing crisis include the *Rebuilding Ireland Action Plan*, which targets key issues such as increasing housing supply, improving affordability, and providing social housing. In addition, various planning and zoning reforms have been implemented to streamline construction and incentivize development in high-demand areas [19].

## 2.4 Integration of ABM in Housing Market Analysis

### 2.4.1 Previous ABM Studies on Housing Markets

Agent-Based Models (ABMs) have been increasingly used to study housing markets due to their ability to capture the complexity of interactions between agents such as households, developers and financial institutions. Notable studies include:

- **Washington D.C. Housing Market:** Geanakoplos et al. (2012) and Axtell et al. (2014) studied the 2007 housing bubble, using ABM to analyze micro-level interactions and the impact of mortgage defaults on market stability [20, 21].

- **British Housing Market:** Baptista et al. (2016) developed an ABM that incorporated a dual housing market, including both buying and renting, and examined the impact of macroprudential policies [22].
- **Dutch Housing Market:** Kerkhoven (2022) modeled the effects of policies targeting reductions in per capita housing space, addressing overcrowding and affordability issues during a housing crisis [23].

These studies highlight the flexibility of ABM in analyzing various housing market phenomena, from price dynamics to the effects of government interventions.

### 2.4.2 Key Components of Housing Market ABMs

Housing market ABMs typically include the following components:

- **Households:** Agents characterized by attributes such as age, household size, income, wealth, and housing preferences.
- **Houses:** Represented as stationary agents with properties like size, quality, and type (e.g., for sale, for rent, or social housing).
- **Markets:** Mechanisms for listing, buying, and renting homes, driven by interactions between agents.
- **Financial Institutions:** Proto-agents providing mortgages and influencing affordability through lending policies.
- **Government:** A proto-agent implementing policies such as taxes, subsidies, or housing programs.
- **Construction Sector:** Mechanisms for building new housing and expanding supply.

### 2.4.3 Modeling Housing Market Dynamics

ABMs allow for the exploration of various housing market dynamics, including:

- **Price Formation:** Emerging from the interactions between buyers, sellers, and renters.
- **Housing Choice:** Modeled through utility functions that consider factors such as affordability, space, and location preferences.
- **Market Segmentation:** The division between different housing sectors, such as ownership, private rental, and social housing.
- **Policy Impacts:** Simulating the effects of government interventions like zoning laws, taxes, and housing subsidies.
- **Demographic Changes:** Capturing household formation, migration, and dissolution over time.

#### 2.4.4 ABM Potential for Irish Housing Analysis

Agent-Based Modeling holds significant potential for analyzing the Irish housing market:

1. **Policy Evaluation:** ABMs can simulate the impacts of proposed policies, such as interventions aimed at improving affordability or increasing housing supply.
2. **Market Segmentation:** Modeling the different housing sectors (ownership, rental, social housing) can provide insights into sector-specific dynamics.
3. **Spatial Dynamics:** ABMs can incorporate geographical factors, crucial for understanding regional variations in housing demand and supply across Ireland.
4. **Long-Term Projections:** ABMs can simulate market trends over extended periods, offering insights into long-term housing affordability and supply-demand balance.

5. **Stakeholder Interactions:** ABMs can model interactions between key stakeholders, such as households, banks, developers, and government agencies.

### 2.4.5 Challenges and Considerations

Despite its advantages, ABM poses several challenges for housing market analysis:

- **Data Requirements:** Accurate micro-level data on households, housing stock, and transactions are essential for model calibration.
- **Validation:** Ensuring the model's output aligns with real-world dynamics can be difficult, particularly for long-term simulations.
- **Computational Complexity:** Large-scale models can be computationally intensive, requiring significant resources for processing.
- **Complexity vs. Usability:** Striking the right balance between model complexity and interpretability for policymakers is crucial.

By addressing these challenges, ABM can provide valuable insights into the complex dynamics of the Irish housing market and support evidence-based policy-making.

# Chapter 3

## Literature Review

This literature review chapter provides a comprehensive overview of agent-based modeling (ABM) applications in housing market analysis. It covers the foundations of ABM in housing markets, examines pioneering and comprehensive market models, discusses the modeling of human behavior and spatial considerations, explores construction and market dynamics, addresses social housing and policy implications, and finally touches on model design, validation, and challenges in ABM research for housing markets.

### 3.1 Introduction

Agent-based modeling has emerged as a powerful tool for simulating complex systems like housing markets [2, 9, 10]. This approach allows for the modeling of heterogeneous agents, their interactions, and the emergence of macro-level phenomena from micro-level behaviors [4]. This chapter reviews key agent-based models of housing markets, their methodologies, and contributions to the field.

## 3.2 Foundations of Agent-Based Modeling in Housing Markets

The application of ABMs to housing markets represents a significant shift from traditional economic models. While classical approaches often rely on equilibrium assumptions and representative agents, ABMs can capture the heterogeneity of market participants and the dynamic, out-of-equilibrium processes that characterize real housing markets [5].

### 3.2.1 Pioneering Studies

The foundational studies in agent-based modeling of housing markets are those by Geanakoplos et al. [20] and Axtell et al. [21]. These studies focused on the Washington D.C. housing market during the 2007 housing bubble and its aftermath. Their key innovation was the use of micro-level empirical calibration, a technique that calibrates agent behavior based on real-world transaction data.

Geanakoplos et al. [20] developed a model that included heterogeneous agents with differing beliefs about future house prices. This heterogeneity was crucial in generating realistic price dynamics and bubble formation. Axtell et al. [21] extended this work, creating a model of the entire Washington D.C. housing market with over 2 million agents. This model was able to reproduce the boom-bust cycle observed in the real market, demonstrating the power of ABMs in capturing complex market dynamics.

## 3.3 Comprehensive Market Models

Building on these foundations, researchers have developed increasingly sophisticated agent-based models (ABMs) of housing markets, each offering unique insights and methodologies that deepen our understanding of housing dynamics.

### 3.3.1 UK Housing Market Model

Baptista et al. [22] developed a comprehensive agent-based model of the UK housing market, which is particularly notable for its dual focus on both the rental and buying markets. This model reflects the complex interactions between these two sectors and includes additional key components such as credit constraints and a simulated banking sector. The incorporation of macroprudential policies into the model allows for the evaluation of regulatory interventions, providing insights into how such policies can mitigate housing market volatility and financial instability.

### 3.3.2 Dutch Housing Market Model

Kerkhoven’s [23] agent-based model of the Dutch housing market makes a significant contribution by focusing on space utilization and policy interventions aimed at reducing space per person. The key features of this model include:

- A detailed representation of households, capturing variations in age, size, income, and wealth.
- Integration of both the buying and rental markets, with a specific emphasis on the social housing sector.
- Utility-based decision-making for households, grounded in the principle of utility maximization.
- Exploration of policy interventions, such as taxing housing space per square meter ( $\text{m}^2$ ) per person and allowing house sharing.
- A demographic model that accounts for household formation and dissolution, although it simplifies population dynamics by excluding explicit modeling of births, deaths, and migration.

Kerkhoven’s model is particularly valuable for its focus on space efficiency, an area often neglected in other housing market models. Additionally, its

detailed treatment of the social housing sector provides crucial insights into how housing policies can influence market dynamics and the distribution of living space, especially in markets where space constraints are a significant concern.

### **3.3.3 High-Resolution Data-Driven Model**

Mér"o et al. [24] recently introduced a high-resolution, data-driven agent-based model of the housing market, offering a more granular perspective on market dynamics. This model employs empirical micro-level data to calibrate a detailed representation of both households and housing stock. One of the key features of this model is its comprehensive population component, which includes mechanisms for births, deaths, and household formation processes. While the model does not explicitly simulate migration, it uses a balancing mechanism to account for population changes, ensuring demographic accuracy.

The high-resolution nature of this model allows for a fine-grained analysis of housing dynamics, making it particularly useful for assessing the impact of localized policies and demographic changes on housing supply and demand.

### **3.3.4 Synthesis and Contribution**

Each of these models makes a unique contribution to the understanding of housing markets:

Together, these models demonstrate the versatility of ABMs in capturing different aspects of housing market dynamics, from macroprudential policies and space efficiency to high-resolution data integration. They also highlight how ABMs can be tailored to address specific research questions and policy concerns, offering valuable tools for both theoretical exploration and practical policy analysis.

| Study                  | Key Focus and Contribution                                                                                                                                                               |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Baptista et al.</b> | Focuses on the interaction between rental and buying markets under macroprudential policies, highlighting the importance of financial regulations in stabilizing housing markets.        |
| <b>Kerkhoven</b>       | Emphasizes space utilization and provides detailed insights into the social housing sector, making it especially relevant for markets facing housing shortages and space inefficiencies. |
| <b>Mér"o et al.</b>    | Delivers a high-resolution, data-driven approach, offering detailed population dynamics that enhance the realism and predictive capacity of housing market models.                       |

Table 3.1: Summary of Key Contributions from Different Housing Market Models

## 3.4 Modeling Human Behavior in Housing Markets

A critical aspect of ABMs in housing markets is the representation of human decision-making processes.

### 3.4.1 Utility-Based Decision Making

Many models use utility-based decision making for agents, grounded in utility maximization theory [5]. This approach assumes that agents choose options that maximize their utility, although there are recognized limitations to human information processing that need to be considered.

### 3.4.2 Behavioral Economics in ABMs

Some researchers have incorporated insights from behavioral economics into their models. For instance, Kerkhoven [23] explored the Dutch housing market, focusing on policies that reduce space per person as a potential solution to the housing crisis. This study demonstrates the importance of considering country-specific factors and behavioral aspects in housing market models.

## **3.5 Spatial Considerations in Housing Market ABMs**

The spatial dimension of housing markets is a crucial factor that ABMs are uniquely positioned to address.

### **3.5.1 Importance of Location**

Traditional economic literature has long recognized the importance of spatial factors in housing markets [14, 15]. ABMs can incorporate these spatial elements more naturally than traditional models, allowing for the emergence of phenomena like neighborhood effects and price spillovers.

### **3.5.2 Challenges in Spatial Modeling**

While many models do not capture the full spatial dimension of housing markets due to increased complexity, some researchers have attempted to address this. Challenges include representing the physical layout of cities, modeling transportation networks, and capturing the influence of amenities and local economic conditions on housing choices.

## **3.6 Construction and Market Dynamics**

The supply side of the housing market, particularly the construction sector, is a critical component of housing market dynamics.

### **3.6.1 Modeling Construction**

Most models implement construction as a proto-agent that balances the number of houses with population size. However, this simplification may not capture the complexities of real-world construction processes, including delays, zoning regulations, and the strategic behavior of developers.

### **3.6.2 Supply-Demand Mismatch**

Recent studies like Curran and Horgan-Jones [1] highlight the mismatch between population growth and housing construction in countries like Ireland, emphasizing the need for more sophisticated modeling of construction dynamics in ABMs.

## **3.7 Social Housing and Policy Implications**

A notable gap in the literature is the detailed modeling of different rental sectors, particularly social housing.

### **3.7.1 Importance of Social Housing**

This is especially important in countries with significant social housing sectors, such as Ireland [17, 18]. ABMs that incorporate social housing can provide insights into the interactions between different housing sectors and the impact of housing policies on different segments of the population.

### **3.7.2 Policy Modeling**

Government policies, such as Ireland’s ”Rebuilding Ireland” plan [19], can have significant impacts on housing markets. ABMs offer a platform for testing the potential effects of such policies before implementation, providing valuable insights for policymakers.

## **3.8 Model Design, Validation, and Challenges**

The design and validation of agent-based models are crucial aspects of the modeling process.

### **3.8.1 Design Considerations**

Abdou et al. [3] provide guidance on designing and building agent-based models, emphasizing the importance of clear objectives, appropriate abstraction levels, and thorough testing.

### **3.8.2 Validation Approaches**

Windrum et al. [7] discuss various approaches to empirical validation of ABMs, including indirect calibration, the Werker-Brenner approach, and the history-friendly approach. Validation remains a challenge in ABM research, particularly for complex systems like housing markets where controlled experiments are not feasible.

### **3.8.3 Computational Challenges**

As models become more complex and data-intensive, computational challenges emerge. Tools like NetLogo [13] have made it easier to implement and experiment with agent-based models, but large-scale, high-resolution models may require more sophisticated computational resources.

# Chapter 4

## Model Development and Methodology

This chapter details an agent-based model simulating the Irish housing market. It describes the model's key components: households, houses, banks, and construction companies, each with specific properties and behaviors. The simulation process, running on monthly and yearly cycles, includes mechanisms for price updates, housing decisions, construction, and demographic changes. The model incorporates various parameters and assumptions related to household behavior, market dynamics, and economic factors. It uses demographic and income data from Ireland's Central Statistics Office to initialize and calibrate the simulation.

### 4.1 Model Overview

The agent-based model (ABM) simulates the Irish housing market, capturing the complex interactions between households, banks, and construction companies within the context of Ireland's unique housing challenges. The model is implemented in C# and is designed to explore various scenarios and potential policy interventions.

The key components of the model include:

- A population of heterogeneous households with varying demographics

and economic characteristics.

- Banks that provide mortgages and influence market liquidity.
- A construction company that build new houses.
- A housing stock consisting of diverse properties.
- Economic factors such as interest rates and income distributions.
- Government policies that can be adjusted to test different interventions.

The model simulates the housing market over a specified time period, typically several years, with each time step representing one month. During each time step, agents make decisions based on their individual characteristics and the current market conditions, leading to emergent market behavior.

## 4.2 Agents and Their Characteristics

### 4.2.1 Households

#### 4.2.1.1 Properties

Households are the primary agents in the model. Each household is characterized by several key properties:

- **Id:** A unique identifier for each household.
- **Members:** A list of Person objects representing the individuals in the household.
- **IncomePercentile:** A value between 0 and 100, representing the household's position in the income distribution.
- **WaitlistTime:** The time spent waiting for social housing, measured in months.
- **Contract:** Represents the household's current housing situation (e.g., owning, renting, or homeless).

- WantToMove: A boolean indicator of whether the household desires to change its housing situation.
- LoanAmount: The current mortgage loan amount, if applicable.
- LoanTermYears: The remaining years on the mortgage loan, if applicable.

#### 4.2.1.2 Initialization

The initialization of households is based on real-world demographic data, ensuring that the model starts with a realistic distribution of household types. The process is implemented in the HouseholdGenerator class and works as follows:

1. The process starts with a dataset (HouseholdDataStore) that provides counts for different household types, including:
  - One Person households
  - Married Couples without children
  - Married Couples with children
  - Single Mother households
  - Single Father households
2. For each household type, the count is scaled down by the model's scale factor to determine how many households of that type to generate.
3. For each household to be created:
  - (a) The household type determines the basic structure (e.g., number of adults, presence of children).
  - (b) For adult members:
    - A random age is selected from the appropriate age distribution (19-24, 25-64, or over 65).

- Gender is assigned based on the household type (e.g., male for single fathers).
  - Income is calculated using the IncomeCalculator, based on the person’s age and gender.
- (c) For households with children:
- The number of children is determined based on a distribution of family sizes.
  - Each child’s age is randomly generated, ensuring it’s appropriate given the parents’ ages.
  - Children are assigned a random gender.
- (d) A unique ID is assigned to each household.
4. After all households are created, income percentiles are calculated:
- Households are sorted by their total income.
  - Each household’s percentile rank is calculated based on its position in this sorted list.
5. Initial housing contracts are assigned to households:
- The type of contract (e.g., owner-occupied, rented, social housing) is determined probabilistically based on real-world housing type distributions.
  - Initially, these contracts are not linked to specific houses.

This initialization process ensures that the starting population closely mirrors the real-world distribution of household types, ages, sizes, and economic situations.

#### **4.2.1.3 Decision-making processes**

The primary decision-making process for households revolves around housing choices. This process is implemented in the ‘DecideToMove’ method and is driven by utility calculations and probabilistic decision-making. The procedure works as follows:

1. Each month, a household evaluates its current housing situation and potential alternatives.
2. The household receives a list of suitable houses, representing available options in the market.
3. For each housing option, including the current residence, the household calculates a utility score using a utility function (detailed below).
4. The household identifies the option with the highest utility.
5. Instead of always choosing the highest utility option, the model calculates a probability of moving:
  - This probability is based on the percentage improvement in utility compared to the current situation.
  - The probability is calculated as:

$$\min \left( 1, \frac{\text{bestUtility} - \text{currentUtility}}{\text{currentUtility}} \times 3 \right) \quad (4.1)$$

- This formula ensures the probability is capped at 100% to remain a valid probability.
6. A random number is generated to determine if the household actually decides to move, based on this probability.
  7. If the decision to move is made, the household's 'WantToMove' flag is set to true.

**Utility Function** The utility function quantifies the desirability of a housing option for a household. It consists of both deterministic and stochastic components:

$$U = U_d + U_s \quad (4.2)$$

Where  $U$  is the total utility,  $U_d$  is the deterministic component, and  $U_s$  is the stochastic component.

The deterministic component  $U_d$  is calculated as:

$$U_d = 0.2 \times \frac{S}{N} \times Q - M - \frac{P}{5Y} \quad (4.3)$$

Where:

- $S$  is the size of the house in square meters
- $N$  is the number of people in the household
- $Q$  is the quality of the house (an integer from 1 to 10)
- $M$  is the movement cost (a barrier to frequent moves)
- $P$  is the price of the house
- $Y$  is the total annual income of the household

The stochastic component  $U_s$  is drawn from a normal distribution:

$$U_s \sim \mathcal{N}(0, 2) \quad (4.4)$$

For the case of homelessness, the utility is set to:

$$U_{homeless} = -10 + U_s \quad (4.5)$$

**Movement Cost** The movement cost  $M$  is calculated using the ‘CalculateMovementCost’ method:

- It starts with a base movement cost.
- If the household has a current contract, the cost is reduced based on how long they’ve been in their current home (up to a 10-year maximum).
- If moving to a different type of housing (e.g., from buying to renting), an additional cost is applied.

This creates a preference for staying in the current home and discourages frequent moves or drastic changes in housing type.

#### 4.2.1.4 Wealth and Income Updates

The model includes mechanisms for updating both the wealth and income of households over time, implemented in the `SimulationUpdater` class.

**Monthly Wealth Updates** Wealth updates occur monthly as part of the ‘`PerformMonthlyUpdates`’ method:

- Monthly housing costs are calculated based on the current contract.
- Monthly savings are calculated as:

$$Savings = TotalIncome - HousingCosts - EssentialExpenses - Taxes \quad (4.6)$$

- Positive savings are added to wealth; negative savings are subtracted.
- If wealth becomes negative, the household is flagged as homeless and their housing contract is terminated.

**Yearly Income Updates** Income updates occur annually in the ‘`PerformYearlyUpdates`’ method:

- For each household member, ‘`UpdateIncome`’ is called.
- New income is calculated using ‘`GetIncome(Age, Gender)`’.
- This reflects changes due to aging and career progression.

#### Additional Methods

- `ResetWaitlistTime()`: Resets the waitlist time to zero, typically called when a household is assigned social housing.
- `IsInterestedInSocialHousing()`: Determines if a household is eligible for and interested in social housing, based on their current contract status and income.
- `IncrementWaitlistTime()`: Increases the waitlist time by one month for households waiting for social housing.

#### 4.2.1.5 Limitations and Assumptions

The current implementation of households carries several important limitations and assumptions:

- **Static Household Composition:** Household structures remain static throughout the simulation.
- **Simplified Age Progression:** The model doesn't account for life-stage transitions affecting housing needs.
- **Limited Income Dynamics:** Income updates are based solely on age and gender.
- **Homogeneous Preferences:** All households use the same utility function for housing decisions.
- **Perfect Information:** Households have perfect information about their utility for different housing options.
- **Rational Decision Making:** Decisions are based on utility maximization, not capturing emotional factors.
- **Simplified Wealth Mechanics:** Complex wealth-building mechanisms are not included.
- **Uniform Tax Treatment:** All households are subject to the same tax rate.
- **No Credit History:** Credit scores are not factored into mortgage approvals.
- **Simplified Moving Costs:** The full complexity of relocation expenses is not captured.
- **Limited Social Housing Dynamics:** Waitlist time doesn't significantly influence allocation beyond tie-breaking.

- Fixed Essential Expenses: A fixed percentage of income is assumed for essential expenses.
- No Retirement Modeling: The model doesn't account for significant changes in income and spending at retirement.
- Limited Homelessness Dynamics: The process of recovering from homelessness is not modeled in detail.
- No Wealth Shocks: Sudden changes in wealth (e.g., inheritances, major expenses) are not modeled.

These simplifications allow for a computationally efficient model while still capturing key aspects of household dynamics in the housing market. Future iterations could incorporate more detailed mechanisms to enhance realism and predictive power.

## **4.2.2 Houses**

### **4.2.2.1 Properties**

Houses are stationary agents in the model, representing the physical dwellings that households can occupy. Each house has the following key properties:

- Size: A numerical value representing the floor area of the house in square meters.
- Quality: An integer score from 1 to 10, representing the overall condition and desirability of the house.
- Type: Indicates whether the house is for social rent, private rent, or buying (represented by the `HouseType` enum: `SocialRent`, `PrivateRent`, `Buy`).
- Owner: For houses that can be owned, this indicates which household currently owns the house.
- Price: The current price or rent of the house.

#### **4.2.2.2 Initialization and Construction**

The initialization and ongoing construction of houses are managed by the `ConstructionCompany` class:

##### **Initial Housing Stock**

1. A dataset provides counts of houses in different size ranges or 'slots'.
2. Counts are scaled down by the model's scale factor.
3. For each house:
  - Size is selected from a normal distribution with mean 150 and standard deviation 50.
  - Quality is assigned from a uniform distribution between 1 and 10.
  - Type (rent or buy) is determined based on a pre-set percentage.

##### **Ongoing Construction**

1. New houses are added annually based on a base construction rate, adjusted for:
  - Current vacancy rate (target is 5%)
  - Market fluctuations (modeled with a random factor)
2. For each new house:
  - Size is drawn from a predefined distribution.
  - Quality is randomly assigned (1-10).
  - Type is determined probabilistically.
  - A check ensures the house meets minimum size requirements for households.

#### 4.2.2.3 Key Methods and Processes

**House Management** Houses have several methods that manage their state and interactions:

- `ChangeOwnership(Household newOwner)`: Updates the owner of the house.
- `UpdatePrice(double factor)`: Adjusts the price of the house based on market conditions.
- `IsOwned()`: Checks if the house has an owner.
- `IsAvailable()`: Determines if the house is available for occupation.

**Housing Contracts** The occupation of houses is managed through `HousingContracts`, which define the relationship between a household and a house:

- **Contract Types**: `OwnerOccupiedWithoutLoan`, `OwnerOccupiedWithLoan`, `RentedFromLandlord`, and `SocialHousing`.
- **Contract Properties**: Include the type, duration (`RunTimeMonths`), remaining mortgage (if applicable), monthly cost, and associated house.
- **Contract Updates**: Contracts are updated monthly, potentially changing type (e.g., when a mortgage is paid off).

**Pricing Mechanisms** The pricing of houses in the model is determined by initial pricing at construction and subsequent monthly adjustments:

1. Initial Pricing:

- For new construction, the initial price is calculated based on the house's size, quality, and type:

$$Price = BasePrice \times Size \times QualityMultiplier \times TypeMultiplier \quad (4.7)$$

Where:

- BasePrice is set to 2000 (presumably currency units per square meter)
- Size is the floor area of the house in square meters
- QualityMultiplier is calculated as  $0.5 + (\text{quality} * 0.1)$ , where quality is an integer from 1 to 10
- TypeMultiplier is 1.2 for houses to buy, and 1.0 for rental properties

## 2. Monthly Price Adjustments:

- Price adjustments occur monthly as part of the SimulationUpdater's PerformMonthlyUpdates method.
- The UpdateHousePrices method is called, which iterates through all houses and updates their prices.
- For each house, a random price change between -5% and 5% is generated.
- The UpdatePrice method of the House class is then called with the factor  $(1 + \text{priceChange})$ .
- This results in each house's price randomly increasing or decreasing by up to 5% each month.

## 3. Market Categorization:

- Houses are categorized into different types (Buy, PrivateRent, SocialRent) upon creation.
- In the ConstructionCompany class, the type is determined probabilistically:
  - 60% chance for Buy
  - 30% chance for PrivateRent
  - 10% chance for SocialRent

This pricing system establishes initial house values based on their characteristics and implements monthly price fluctuations to simulate market

dynamics. The initial pricing reflects the inherent value of the house, while the monthly updates introduce an element of market volatility.

#### **4.2.2.4 Limitations and Assumptions**

The current implementation of houses in the model carries several important limitations and assumptions:

- **Homogeneous Quality Metrics:** The quality score doesn't distinguish between different aspects of house quality (e.g., location, amenities).
- **Simplified Pricing Model:** The pricing model doesn't account for complex factors like neighborhood effects or economic cycles.
- **Static House Characteristics:** Houses don't age or depreciate over time, and renovations are not modeled.
- **Uniform Construction:** New houses are added without considering geographical constraints or zoning laws.
- **Instantaneous Construction:** New houses become immediately available, not accounting for construction time.
- **Limited Market Dynamics:** The model might not fully capture the complexities of real estate market cycles.
- **Simplified Rental Market:** The distinction between private and social rental is based solely on price, which may oversimplify real-world dynamics.

These simplifications allow for a computationally efficient model while still capturing key aspects of the housing market. Future iterations could incorporate more detailed mechanisms to enhance realism and predictive power.

### 4.2.3 Bank

#### 4.2.3.1 Properties

The Bank in the model represents a financial institution that provides mortgages to households. It has the following key properties:

- **MortgagedHouseholds:** A list of households with active mortgages.
- **InterestRate:** The current interest rate applied to mortgages.
- **MaxLoanToValueRatio:** A constant set to 0.9 (90% LTV), representing the maximum allowed loan-to-value ratio.
- **MaxDebtToIncomeRatio:** A constant set to 3.0, representing the maximum allowed debt-to-income ratio.

#### 4.2.3.2 Initialization

The Bank is initialized with an initial interest rate:

1. The **MortgagedHouseholds** list is created as an empty list.
2. The **InterestRate** is set to the provided initial rate.

#### 4.2.3.3 Key Methods and Processes

**Mortgage Approval** The **ApproveMortgage** method determines whether a mortgage application is approved:

1. Calculate the loan amount and loan-to-value ratio.
2. Check if the loan-to-value ratio exceeds the maximum allowed (90%).
3. Calculate the monthly payment using the **CalculateMonthlyPayment** method.
4. Check if the annual mortgage payments exceed one-third of the household's annual income.

5. If approved:

- Update the household's loan amount and term.
- Add the household to the MortgagedHouseholds list.

6. Return true if approved, false otherwise.

**Payment Collection** The CollectPayments method manages monthly mortgage payments:

1. Iterate through all mortgaged households.

2. For each household:

- Calculate the monthly payment, interest, and principal components.
- If the household can make the payment:
  - Reduce the loan amount by the principal payment.
  - If the loan is fully paid, remove the household from MortgagedHouseholds.
- If the household cannot make the payment:
  - Mark the household as defaulted.
  - Remove the household from MortgagedHouseholds.

**Interest Rate Update** The UpdateInterestRate method allows for changing the interest rate:

- Accept a new interest rate as input.
- Validate that the new rate is not negative.
- Update the InterestRate property.

#### **4.2.3.4 Limitations and Assumptions**

The current implementation of the Bank in the model carries several important limitations and assumptions:

- **Fixed Interest Rate:** The model uses a single, adjustable interest rate for all mortgages, not reflecting the variety of rate types in real markets.
- **Simplified Approval Criteria:** Mortgage approval is based only on loan-to-value and debt-to-income ratios, omitting other factors like credit scores.
- **Homogeneous Mortgage Products:** All mortgages are treated as fixed-rate loans with the same terms, not capturing the diversity of real-world mortgage products.
- **Limited Default Handling:** While the model identifies defaults, it does not implement specific foreclosure procedures.
- **Unlimited Lending Capacity:** The model assumes the bank has unlimited funds to provide mortgages, not considering the bank's capital or overall financial health.
- **Simplified Payment Structure:** The model doesn't account for factors like property taxes or insurance in mortgage payments.

These simplifications allow for a computationally efficient model while still capturing key aspects of mortgage lending in the housing market. Future iterations could incorporate more detailed financial mechanisms to enhance realism and predictive power.

### **4.2.4 Construction Company**

#### **4.2.4.1 Properties**

The `ConstructionCompany` in the model represents the entity responsible for adding new houses to the market. It has the following key properties:

- `baseAnnualNewHouseCount`: A constant set to 20000, representing the base number of houses to construct annually.
- `scaleFactor`: A factor used to scale down the construction numbers for the simulation.
- `minimumAreaPerPerson`: A constant set to 20.0, representing the minimum area per person for a house to be considered suitable.
- `houseConstructed`: An event that is triggered when a new house is constructed.

#### 4.2.4.2 Initialization

The `ConstructionCompany` is initialized with a scale factor:

1. The `scaleFactor` is set to the provided value.

#### 4.2.4.3 Key Methods and Processes

**Construct New Houses** The `ConstructNewHouses` method is the primary function for adding new houses to the market:

1. Calculate the current vacancy rate in the housing market.
2. Determine the base number of houses to construct, adjusted by the scale factor.
3. Adjust the construction count based on the difference between current and target vacancy rates.
4. Apply a random factor (between 0.8 and 1.2) to simulate market fluctuations.
5. For each new house to be constructed:
  - Determine the size using `DrawSizeFromDistribution` method.
  - Assign quality using `DrawQualityFromDistribution` method.

- Determine the house type (Buy, PrivateRent, or SocialRent) using DetermineHouseType method.
- Calculate the initial price using CalculateInitialPrice method.
- Check if the house is suitable for the market using HouseIsSuitableForMarket method.
- If suitable, add the house to the market and trigger the HouseConstructed event.

### House Property Generation

- Size: Drawn from a normal distribution with mean 150 and standard deviation 50, clamped between 50 and 300 square meters.
- Quality: Uniformly distributed between 1 and 10.
- Type: Probabilistically determined (60% Buy, 30% PrivateRent, 10% SocialRent).

**Initial Price Calculation** The initial price of a new house is calculated based on its properties:

$$Price = BasePrice \times QualityMultiplier \times TypeMultiplier \quad (4.8)$$

Where:

- $basePrice = Size * 2000$
- $qualityMultiplier = 0.5 + (Quality * 0.1)$
- $typeMultiplier = 1.2$  for Buy,  $1.0$  for Rent types

#### 4.2.4.4 Market Dynamics

The ConstructionCompany simulates several market dynamics:

- **Vacancy Rate Targeting:** Adjusts construction to maintain a 5% vacancy rate.
- **Market Fluctuations:** Incorporates randomness in construction numbers to simulate unpredictability.
- **Supply Segmentation:** Probabilistically determines the type of each new house (Buy, PrivateRent, SocialRent).
- **Market Suitability:** Ensures new houses meet minimum size requirements based on household sizes in the market.

#### **4.2.4.5 Limitations and Assumptions**

The current implementation of the ConstructionCompany in the model carries several important limitations and assumptions:

- **Fixed Base Construction Rate:** The base annual construction rate is constant and may not reflect real-world variability in construction capacity.
- **Simplified Size Distribution:** House sizes are based on a normal distribution, which may not capture the full complexity of real-world housing stock.
- **Uniform Quality Distribution:** The quality of new houses is uniformly distributed, which might not reflect real-world tendencies in new construction.
- **Instant Construction:** New houses are immediately available, not accounting for construction time or phased developments.
- **Location Agnostic:** The model doesn't account for the location of new constructions, which can be a significant factor in real-world housing markets.

- **Simplified Price Calculation:** The pricing model for new houses is based on simple factors and doesn't account for complex market dynamics or location value.
- **Limited Constraint Modeling:** The model doesn't account for constraints like land availability, zoning laws, or construction resource limitations.

These simplifications allow for a computationally efficient model while still capturing key aspects of housing supply dynamics. Future iterations could incorporate more detailed mechanisms to enhance realism and predictive power in simulating housing construction and market entry.

## 4.3 Simulation Process

The simulation process in this model is designed to capture the complex dynamics of the Irish housing market over time. It is primarily managed by the `Simulation` and `SimulationUpdater` classes, which work in concert to simulate market behaviors, demographic changes, and economic factors. This section provides a detailed overview of the simulation process, its key components, and the various mechanisms that drive the model.

### 4.3.1 Simulation Overview

The simulation operates on a monthly time step, with certain processes occurring annually. It typically runs for several years (e.g., 10 years) to capture long-term trends in the housing market. The main components of the simulation include:

- **Households:** Representing the population with various attributes such as income, size, and housing preferences.
- **Houses:** Representing the housing stock with attributes like size, quality, and price.

- Bank: Handling mortgage approvals and payments.
- Construction Company: Managing the creation of new houses.
- Economic Factors: Including interest rates, tax rates, and market conditions.

### 4.3.2 Initialization

The simulation begins with an initialization process, implemented in the Simulation class:

#### 4.3.2.1 Initialize(scaleFactor)

This function sets up the initial state of the simulation:

- Generates initial households using the HouseholdGenerator, based on scaled-down Irish demographic data.
- Initializes the housing stock, representing a scaled version of the Irish housing market.
- Sets up economic parameters such as interest rates and tax rates.
- Scales the simulation based on the provided scale factor to manage computational complexity while maintaining representative dynamics.

### 4.3.3 Main Simulation Loop

The core of the simulation is the Run(years, scaleFactor) function in the Simulation class. This function:

- Calls Initialize() to set up the simulation.
- Iterates through each year and month of the simulation period.
- Calls appropriate update functions for monthly and yearly processes.
- Outputs statistics at monthly and yearly intervals for analysis.

### **4.3.4 Monthly Processes**

The simulation performs several crucial updates on a monthly basis, primarily implemented in the `SimulationUpdater` class:

#### **4.3.4.1 Update House Prices**

- Frequency: Monthly for all houses in the simulation.
- Mechanism: Each house's price is adjusted by a small random percentage, typically within  $\pm 5\%$  of the current price.
- Purpose: Simulates market volatility and short-term fluctuations in housing values.

#### **4.3.4.2 Update Contracts**

- Processes monthly mortgage payments for households with mortgages.
- Collects monthly rent for renting households.
- Increments the duration of each contract.
- Handles changes in contract status (e.g., mortgage paid off).

#### **4.3.4.3 Decide To Move**

- Each household evaluates its current housing situation against other available options.
- Calculates a utility score for the current home and potential alternatives.
- Determines whether a household decides to move based on potential improvement in housing utility.

#### **4.3.4.4 Act On Best Options**

- Executes moves for households that have decided to relocate.
- Finds the best available house for each moving household.
- Processes the move, including mortgage approval if necessary.
- Updates the householdMoves counter to track migration within the simulation.

#### **4.3.4.5 Assign Social Housing**

- Identifies households eligible for social housing.
- Assigns available social housing units based on a priority system (wait-list time).
- Updates housing contracts and moves counter accordingly.

#### **4.3.4.6 Process New Listings**

- Simulates houses becoming available due to various factors (e.g., previous owners moving out).
- Updates the availability status of these houses in the simulation.
- Sets initial listing prices for new market entries.

#### **4.3.4.7 Collect Payments**

- Collects mortgage payments from households.
- Updates loan balances and handles any defaults.
- Processes rent collections for rented properties.

### **4.3.5 Yearly Processes**

The simulation performs several important updates on an annual basis:

#### **4.3.5.1 Update Population**

- Applies age and gender-specific mortality rates to the population.
- Simulates new births based on fertility rates of women of childbearing age.
- Adjusts the population based on net migration rates.
- Increments the age of all individuals in the population.

#### **4.3.5.2 Update Incomes**

- Recalculates income for each household member based on updated age and current economic factors.
- Accounts for career progression and general economic trends.

#### **4.3.5.3 Set Tax Rate**

- Calculates the average household income across all households.
- Adjusts the tax rate based on predetermined thresholds:
  - Increases by 1 percentage point if average income exceeds €50,000.
  - Decreases by 1 percentage point if average income falls below €30,000.
- Constrains the tax rate between 10% and 40%.
- Applies the new tax rate to all households, affecting their disposable income and housing decisions.

#### **4.3.5.4 Construct New Houses**

- Determines the number of new houses to build based on current market conditions and the target vacancy rate.
- Creates new houses with varying sizes, qualities, and types (buy, private rent, social rent).

- Adds these new houses to the available housing stock.

### **4.3.6 Data Collection and Analysis**

Throughout the simulation, data is collected and analyzed to track the evolution of the housing market:

#### **4.3.6.1 Output Monthly Statistics**

- Calculates key metrics such as total households, population, average house price, homeownership rate, etc.
- Records these statistics in an Excel worksheet for later analysis.

#### **4.3.6.2 Output Yearly Statistics**

- Similar to monthly statistics, but provides a yearly summary.
- Allows for analysis of long-term trends in the housing market.

#### **4.3.6.3 Calculate Statistics**

- Computes various statistical measures about the current state of the simulation.
- Includes metrics like total households, total population, average household size, vacancy rates, etc.
- Used by both monthly and yearly output functions to ensure consistent calculation of key metrics.

## **4.4 Model Parameters and Assumptions**

This section outlines the key parameters and critical assumptions that form the basis of our Irish housing market simulation model. These elements are fundamental to understanding the scope, limitations, and interpretations of the model's outputs.

### **4.4.1 Key Parameters**

#### **4.4.1.1 Temporal Parameters**

- Simulation Time Step: 1 month
- Simulation Duration: Typically set to several years (e.g., 10 years)
- Update Frequencies: Monthly for market dynamics, yearly for demographic and economic shifts

#### **4.4.1.2 Household Parameters**

- Initial Population Size: Based on scaled-down Irish demographic data.
- Household Types: One Person, Married Couples without children, Married Couples with children, Single Mother households, Single Father households.
- Age Distribution: Derived from Irish demographic data.
- Income Distribution: Based on age and gender, derived from 2020 Irish income statistics.

#### **4.4.1.3 Housing Parameters**

- House Types: Buy, PrivateRent, SocialRent
- House Size Range: 50 to 300 square meters
- House Quality Score: Integer from 1 to 10
- Initial Housing Stock: Scaled representation of Irish housing market
- Minimum Area per Person: 20.0 square meters

#### **4.4.1.4 Economic Parameters**

- Base Annual New House Construction: 20,000 (scaled down in the simulation)
- Interest Rate: Fixed rate set at model initialization
- Maximum Loan-to-Value Ratio: 90%
- Maximum Debt-to-Income Ratio: 3.0
- Tax Rate Range: 10% to 40%, adjusted annually

#### **4.4.1.5 Market Dynamics Parameters**

- Monthly House Price Fluctuation:  $\pm 5\%$
- Target Vacancy Rate: 5%
- Construction Rate Adjustment: Based on current vacancy rate
- Liberalization Threshold: Determines classification between social and private rental housing

### **4.4.2 Critical Assumptions**

#### **4.4.2.1 Household Behavior**

- Households make rational decisions based on utility maximization.
- All households use the same utility function for housing decisions.
- Household composition remains static throughout the simulation.

#### **4.4.2.2 Housing Market**

- House prices are primarily driven by size, quality, and type.
- New construction is immediately available on the market.

- All areas are considered equally desirable (no location-based preferences).
- The market is closed, with no external influences beyond modeled factors.

#### **4.4.2.3 Economic Factors**

- A simplified tax system with a single, adjustable rate applies to all households.
- Bank has unlimited lending capacity.
- All mortgages are fixed-rate loans with the same term.
- Income is solely a function of age and gender.

#### **4.4.2.4 Government Policies**

- Tax rate adjustments are the primary form of government intervention.
- Social housing allocation is based on a simple waiting list system.
- No complex policy interventions such as first-time buyer schemes are modeled.

### **4.4.3 Limitations**

- The model does not account for external economic shocks or major policy changes.
- Geographical factors and regional variations are not considered.
- The simulation does not model the construction industry in detail.
- Complex household formations or dissolutions (e.g., divorces, children moving out) are not simulated.
- The model assumes a closed system with no interaction with other housing markets or economies.

## 4.5 Data and Data Sources

This section provides an overview of the data used in this project, along with their respective sources. All data has been obtained from the Central Statistics Office (CSO) website.

### 4.5.1 Demographic Data

#### 4.5.1.1 Household Types

| Household Type               | Count   |
|------------------------------|---------|
| One Person                   | 425,974 |
| Married Couple               | 354,329 |
| Married Couple With Children | 619,987 |
| One Mother With Children     | 155,583 |
| One Father With Children     | 26,812  |

Table 4.1: Household Type Counts

**Source:** Census of Population 2022 Profile 3 - Households, Families and Childcare (CSO)

#### 4.5.1.2 Children Distribution

| Number of Children | Count   |
|--------------------|---------|
| 1                  | 346,938 |
| 2                  | 323,796 |
| 3                  | 157,160 |
| 4                  | 44,846  |
| 5                  | 9,593   |
| 6                  | 2,392   |
| 7 or more          | 1,174   |

Table 4.2: Distribution of Families by Number of Children

**Source:** Census of Population 2022 Profile 3 - Households, Families and Childcare (CSO)

#### 4.5.1.3 Child Age Groups

| Age Group  | Count   |
|------------|---------|
| Pre-School | 331,515 |
| Primary    | 548,693 |
| Secondary  | 371,588 |

Table 4.3: Child Age Group Counts

**Source:** Census of Population 2022 - Profile 3 An Age Profile of Ireland (CSO)

#### 4.5.1.4 Adult Age Groups

| Age Group | Count     |
|-----------|-----------|
| 19-24     | 331,208   |
| 25-64     | 2,541,294 |
| Over 65   | 637,567   |

Table 4.4: Adult Age Group Counts

**Source:** Census of Population 2022 - Profile 3 An Age Profile of Ireland (CSO)

#### 4.5.1.5 Housing Types

| Contract Type               | Count   |
|-----------------------------|---------|
| Owner Occupied Without Loan | 679,718 |
| Owner Occupied With Loan    | 531,207 |
| Rented From Landlord        | 348,493 |
| Social Housing              | 165,178 |

Table 4.5: Housing Type Counts

**Source:** Census of Population 2022 Profile 2 - Housing in Ireland (CSO)

## 4.5.2 Income Data

### 4.5.2.1 Mean Income by Age Group and Gender (2022)

| Age Group | Male     | Female |
|-----------|----------|--------|
| Overall   | 0.00     | 0.00   |
| 15-24     | 401.66   | 344.92 |
| 25-29     | 692.20   | 628.25 |
| 30-39     | 927.24   | 767.05 |
| 40-49     | 1,106.87 | 825.52 |
| 50-59     | 1,106.31 | 756.63 |
| 60+       | 851.77   | 584.27 |

Table 4.6: Mean Income by Age Group and Gender (2022)

**Source:** Earnings Analysis using Administrative Data Sources 2022 (CSO)

**Note:** Income figures are in EUR per week.

## 4.5.3 Data Notes

- All demographic data is from the Census of Population 2022, conducted by the Central Statistics Office (CSO) of Ireland.
- The income data source and currency details are yet to be specified.
- Age groups in the income data represent the starting age of each group.

# Chapter 5

## Results

This chapter presents results from an agent-based model of the Irish housing market, simulating three scenarios: 3% interest rate, 6% interest rate, and limited construction with restricted vacancy. Key findings show that lower interest rates stimulate higher market activity, while higher rates lead to more stability but less activity. Supply constraints moderate market activity and introduce volatility. Surprisingly, homeownership rates remain stable across scenarios, with the constrained scenario showing the highest rate. The results highlight the complex interplay between interest rates and supply constraints in shaping housing market dynamics, demonstrating the model’s potential for policy analysis and market prediction.

### 5.1 Model Validation

Before presenting the main simulation results, it is important to validate the model’s ability to accurately represent the initial conditions of the Irish housing market. Figures 5.1 and 5.2 illustrates the comparison between the original data and this simulation’s generated housing types.

Figure 5.2 demonstrates that the simulation closely replicates the distribution of contract types observed in the original data, validating the model’s ability to initialize housing tenure accurately.

Table 5.1 shows the distribution of household sizes in the simulated pop-

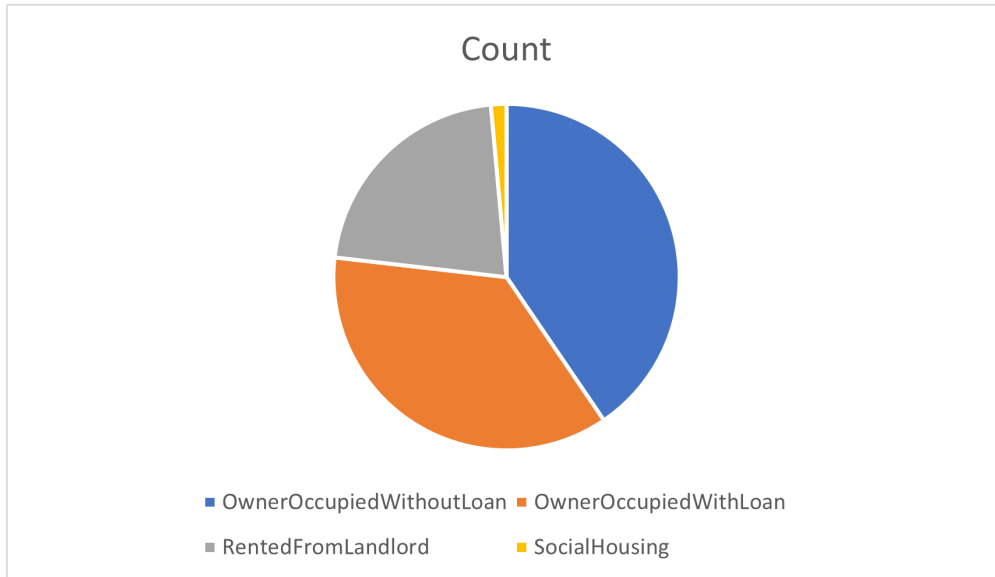


Figure 5.1: Original Data

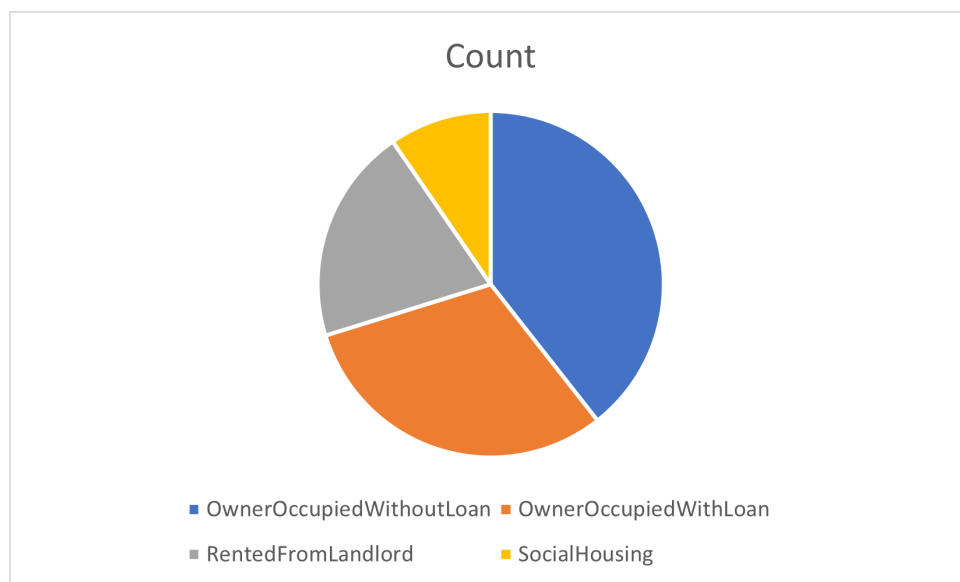


Figure 5.2: Comparison of Contract Types: Original vs. Simulation-Generated Data

Table 5.1: Household Size Distribution

| Household Size | Count |
|----------------|-------|
| 1              | 426   |
| 2              | 424   |
| 3              | 291   |
| 4              | 273   |
| 5              | 115   |
| 6              | 45    |
| 7              | 8     |
| 8              | 1     |

ulation, reflecting a realistic range from single-person households to large families.

Table 5.2 provides a detailed breakdown of household compositions, showing the diversity of family structures captured in the simulation.

Table 5.3 shows the initial distribution of contract types in the simulation, closely mirroring the proportions observed in the Irish housing market.

These comparisons and initial data distributions demonstrate that the simulation accurately represents the diverse household structures and housing tenure arrangements observed in the Irish housing market, providing a solid foundation for subsequent analysis.

## 5.2 Overview of Simulation Outcomes

Three distinct scenarios were created in the simulation: a 3% interest rate scenario, a 6% interest rate scenario, and a scenario with limited construction and restricted vacancy. Figures 5.3, 5.4, and 5.9 illustrate the number of households wanting to move over time for each scenario, respectively. Additionally, Figures 5.5, 5.6, and 5.8 show the homeownership rates for each scenario.

The graphs demonstrate clear differences in household behavior and homeownership trends across the three scenarios. In the 3% scenario (Figure 5.3), a strong upward trend in the number of households wanting to move is observed, reaching a maximum of 1,481. The 6% scenario (Figure 5.4) shows

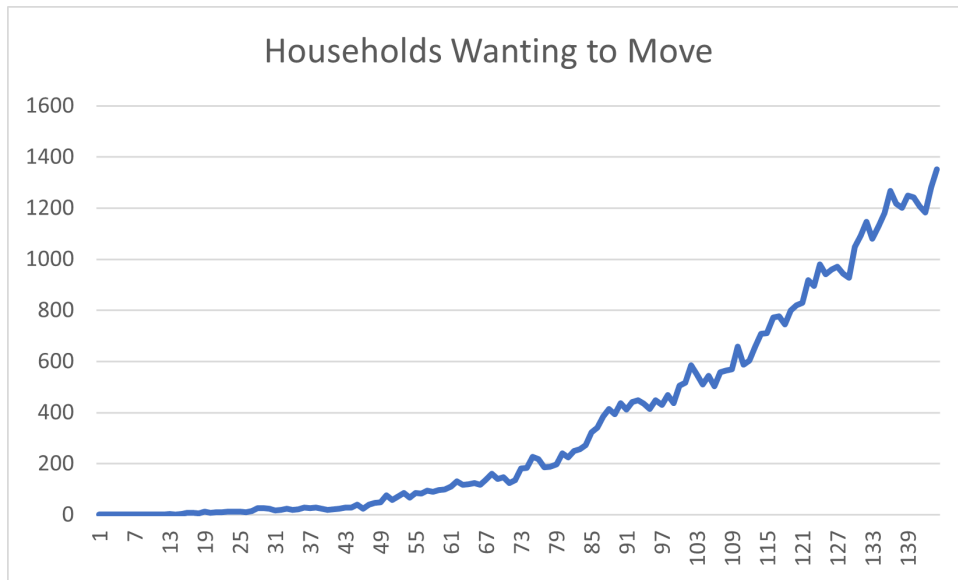


Figure 5.3: Households Wanting to Move Over Time: 3% Interest Rate

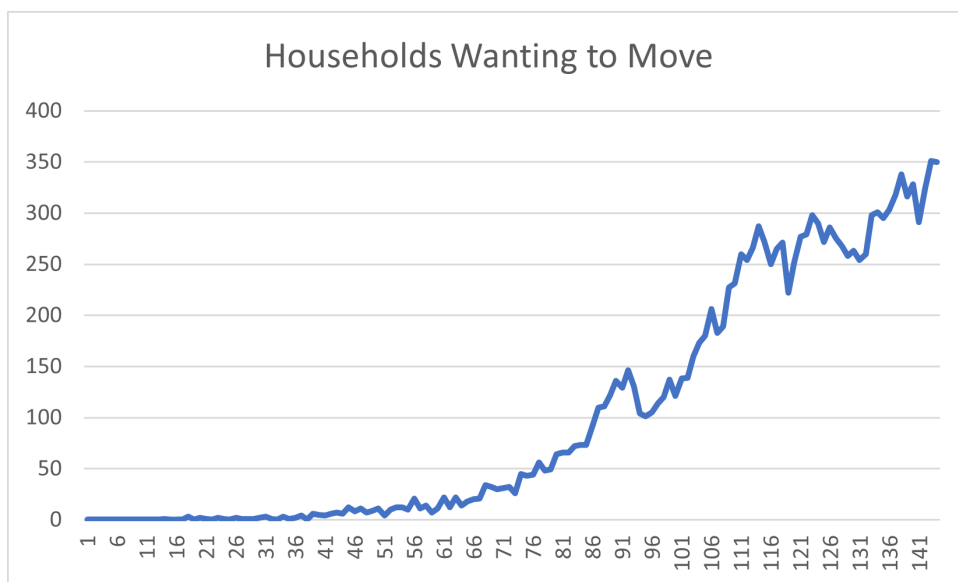


Figure 5.4: Households Wanting to Move Over Time: 6% Interest Rate

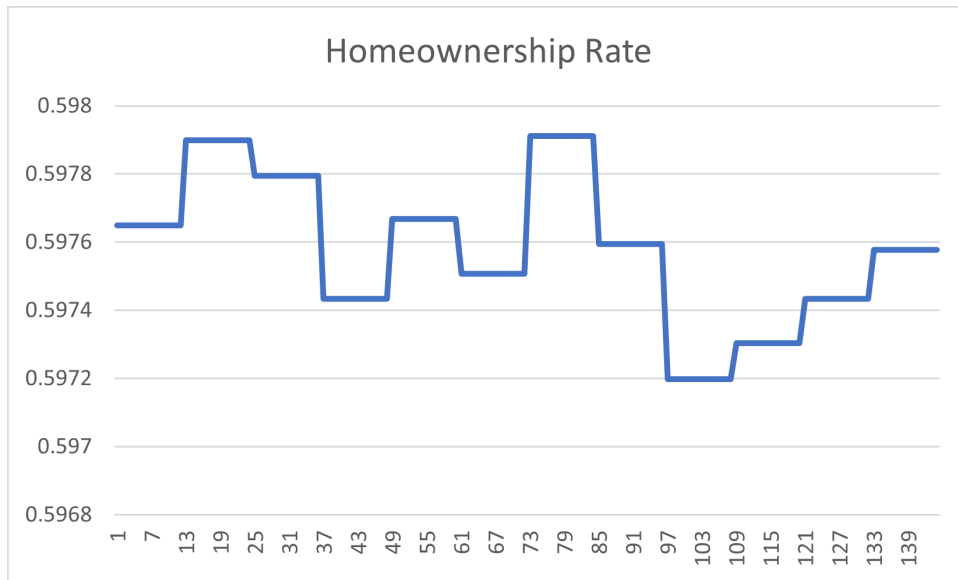


Figure 5.5: Homeownership Rate Over Time: 3% Interest Rate

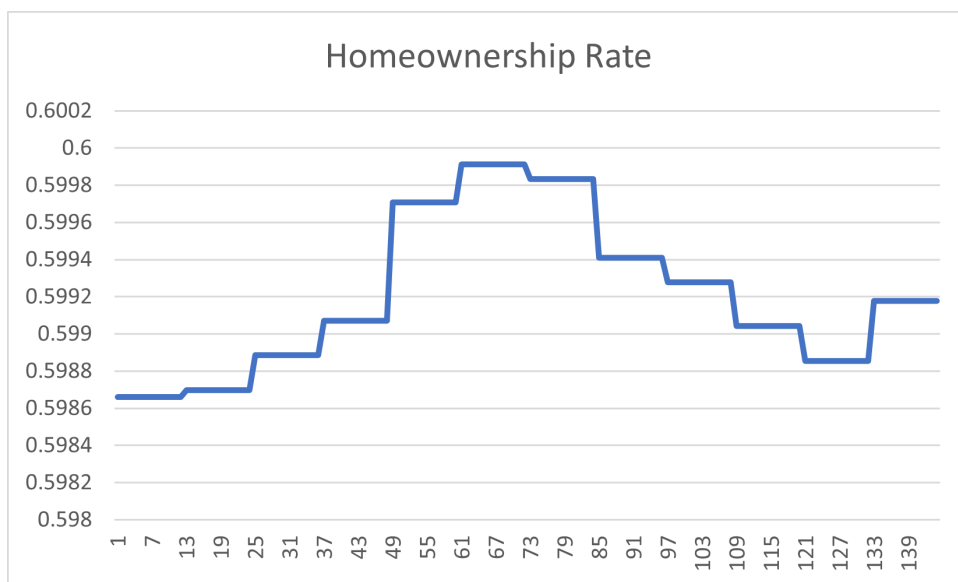


Figure 5.6: Homeownership Rate Over Time: 6% Interest Rate

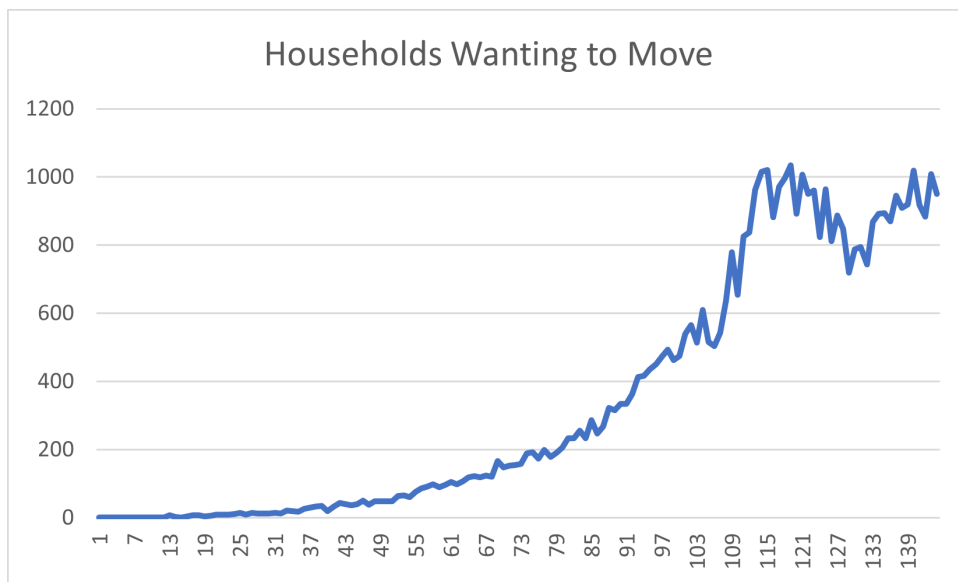


Figure 5.7: Households Wanting to Move Over Time: Limited Construction and Restricted Vacancy

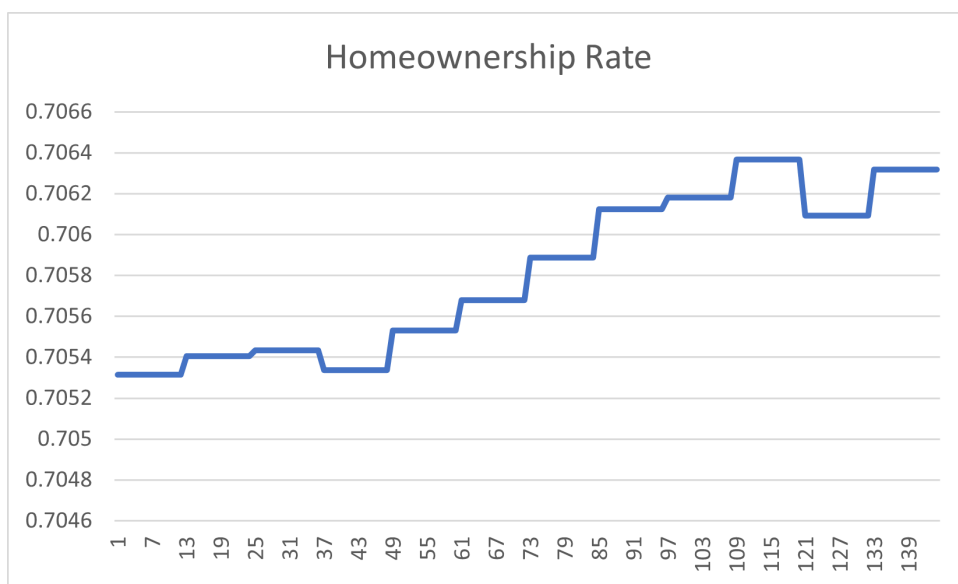


Figure 5.8: Homeownership Rate Over Time: Limited Construction and Restricted Vacancy

Table 5.2: Household Composition: Adults and Children

| Adults | Children | Count |
|--------|----------|-------|
| 1      | 0        | 426   |
| 1      | 1        | 46    |
| 1      | 2        | 21    |
| 1      | 3        | 20    |
| 1      | 4        | 2     |
| 2      | 0        | 378   |
| 2      | 1        | 182   |
| 2      | 2        | 149   |
| 2      | 3        | 59    |
| 2      | 4        | 23    |
| 2      | 5        | 2     |
| 3      | 0        | 88    |
| 3      | 1        | 74    |
| 3      | 2        | 28    |
| 3      | 3        | 9     |
| 3      | 4        | 2     |
| 3      | 5        | 1     |
| 4      | 0        | 30    |
| 4      | 1        | 20    |
| 4      | 2        | 6     |
| 4      | 3        | 2     |
| 5      | 0        | 6     |
| 5      | 1        | 5     |
| 5      | 2        | 2     |
| 6      | 0        | 2     |

a more modest increase, peaking at 351 households. The constrained scenario (Figure 5.9) presents a middle ground, reaching a maximum of 1,034 households wanting to move, but with more pronounced fluctuations.

## 5.3 Key Findings

### 5.3.1 Market Activity

Comparing Figures 5.3 and 5.4, we can observe that the 3% interest rate scenario resulted in significantly higher market activity:

Table 5.3: Initial Distribution of Contract Types

| Contract Type            | Count |
|--------------------------|-------|
| OwnerOccupiedWithoutLoan | 641   |
| OwnerOccupiedWithLoan    | 575   |
| RentedFromLandlord       | 344   |
| SocialHousing            | 23    |

- By the end of the simulation, approximately 4 times more households were wanting to move in the 3% scenario compared to the 6% scenario.
- The rate of increase in households wanting to move was consistently higher in the 3% scenario, as evident from the steeper slope in Figure 5.3.

The constrained scenario shows:

- A peak number of households wanting to move (1,034) that is lower than the 3% scenario but higher than the 6% scenario.
- Greater volatility in household moving decisions, with more frequent and larger fluctuations.
- Periodic declines in the number of households wanting to move, unlike the steadily increasing trends in the unconstrained scenarios.

### 5.3.2 Market Stability

Figure 5.4 demonstrates that the 6% scenario exhibited greater market stability:

- Lower volatility in the number of households wanting to move from month to month, as seen in the smoother curve.
- More frequent periods of zero or very low numbers of households wanting to move, especially in the early stages of the simulation.

The constrained scenario exhibits:

- Higher volatility compared to both unconstrained scenarios, likely reflecting the impact of limited housing availability on household decisions.
- A late-stage plateau and slight decline in households wanting to move, possibly indicating a saturation point under constrained conditions.

### 5.3.3 Long-term Effects

Both figures show an increasing trend over time, but with notable differences:

- Figure 5.3 exhibits a more pronounced increase, suggesting a compounding effect of lower interest rates over time.
- Figure 5.4 shows a more gradual increase with more periods of stagnation or decrease.

### 5.3.4 Homeownership Rates

Analysis of Figures 5.5 and 5.6 reveals interesting trends in homeownership rates:

- **Overall Levels:** The 6% scenario shows slightly higher homeownership rates (around 59.9%) compared to the 3% scenario (around 59.7%).
- **Stability:** Both scenarios demonstrate remarkable stability in homeownership rates over time, with only minor fluctuations.
- **Trend Differences:**
  - 6% Scenario: Shows a slight upward trend, starting at 59.87% and ending at 59.92%.
  - 3% Scenario: Exhibits more variability, with slight decreases in some periods, starting at 59.76% and ending at 59.76%.
- **Variability:** The 3% scenario shows more variability in homeownership rates compared to the 6% scenario, which remains more stable throughout the simulation.

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Analysis of Figure 5.8 for the constrained scenario reveals:

- A significantly higher homeownership rate (around 70.6%) compared to both unconstrained scenarios.
- Remarkable stability throughout the simulation, with only minor fluctuations.
- A very slight upward trend, with the rate increasing from about 70.53% to 70.63% over the course of the simulation.

## 5.4 Interpretation

The simulation results provide several insights into the role of interest rates in shaping housing market dynamics:

1. **Affordability:** Lower interest rates (3%) appear to make moving more affordable, encouraging more households to consider relocation.
2. **Decision Delay:** Higher interest rates (6%) seem to cause households to delay moving decisions, as evidenced by the slower increase in households wanting to move.
3. **Market Segments:** The stark difference between scenarios suggests that higher interest rates particularly affect certain market segments, such as first-time buyers or those looking to upgrade.
4. **Economic Implications:** The 3% scenario indicates a more robust economy with higher housing market activity, while the 6% scenario suggests a more conservative market with potentially slower price growth.
5. **Homeownership Stability:** Despite significant differences in market activity between the two scenarios, homeownership rates remain remarkably stable. This suggests that other factors, such as cultural preferences or long-term housing policies, may have a stronger influence on overall homeownership levels than interest rates alone.

6. **Interest Rate Impact on Homeownership:** Contrary to what might be expected, the higher interest rate scenario (6%) shows slightly higher homeownership rates. This could indicate that higher interest rates might discourage speculative buying or reduce housing price inflation, potentially making homeownership more accessible for some households in the long term.
7. **Market Dynamics vs. Homeownership:** The stark difference in market activity (households wanting to move) between the two scenarios doesn't translate into significant differences in homeownership rates. This suggests that while interest rates strongly influence market liquidity, their impact on the overall distribution of housing tenure types may be more nuanced.
8. **Supply Constraint Effects:** The constrained scenario demonstrates that limitations on construction and vacancy availability can significantly moderate market activity, even in low-interest-rate environments. This highlights the crucial role of housing supply in market dynamics.
9. **Homeownership in Constrained Markets:** Surprisingly, the constrained scenario shows the highest homeownership rate. This suggests that limited housing supply might actually favor existing homeowners, possibly by reducing competition from new entrants to the housing market.
10. **Market Adaptability:** The periodic declines and increased volatility in the constrained scenario indicate that households adapt their behavior based on housing availability, not just financial conditions.
11. **Long-term Market Saturation:** The late-stage plateau in the constrained scenario suggests that sustained supply constraints could lead to a state of reduced mobility in the housing market, potentially impacting broader economic dynamics.

## 5.5 Impact of Limited Construction and Restricted Vacancy

In addition to the initial simulations with 3% and 6% interest rates, a scenario was constructed where construction is limited and vacant houses are not always available. This scenario more closely mimics real-world constraints in the housing market. Figure 5.9 illustrates the number of households wanting to move over time under these conditions.

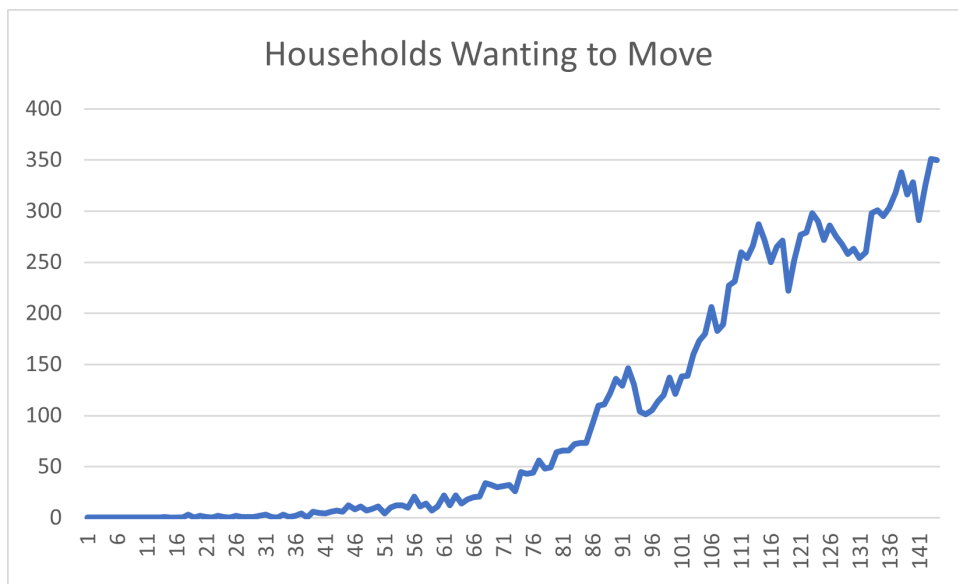


Figure 5.9: Households Wanting to Move Over Time: Limited Construction and Restricted Vacancy

### 5.5.1 Key Observations

Comparing Figure 5.9 with the previous results reveals several important insights:

- **Overall Trend:** The number of households wanting to move still shows an increasing trend over time, but with more pronounced fluctuations compared to the unconstrained scenarios.

- **Maximum Value:** The peak number of households wanting to move reaches 1034, which is lower than the 3% interest rate scenario (1481) but higher than the 6% scenario (351).
- **Volatility:** There is greater volatility in the number of households wanting to move, with more frequent and larger fluctuations. This likely reflects the impact of limited housing availability on household decisions.
- **Periodic Declines:** Unlike the steadily increasing trends in the unconstrained scenarios, this scenario shows periodic declines in the number of households wanting to move. These could represent periods where the lack of available housing discourages households from considering relocation.
- **Late-Stage Behavior:** Towards the end of the simulation, there's a noticeable plateau and even a slight decline in households wanting to move, possibly indicating a saturation point in the market under these constrained conditions.

### 5.5.2 Interpretation

The results from this constrained scenario provide additional insights into housing market dynamics:

1. **Supply Constraints:** The limitations on construction and vacancy availability clearly dampen the overall desire or ability of households to move, as evidenced by the lower peak compared to the 3% unconstrained scenario.
2. **Market Frustration:** The higher volatility and periodic declines in households wanting to move might indicate periods of market frustration, where households desire to move but are unable to find suitable options.

3. **Realistic Market Friction:** This scenario likely represents a more realistic portrayal of housing market friction, where the desire to move is moderated by the practical limitations of housing availability.
4. **Policy Implications:** The results suggest that policies aimed at increasing housing supply and availability could have a significant impact on market liquidity and household mobility.
5. **Long-term Market Dynamics:** The late-stage plateau indicates that under sustained supply constraints, the housing market might reach a state of reduced mobility, potentially leading to broader economic impacts.

### 5.5.3 Comparison with Unconstrained Scenarios

When compared to the 3% and 6% interest rate scenarios:

- The constrained scenario shows a middle ground in terms of overall market activity, suggesting that supply constraints can moderate the stimulating effect of low interest rates.
- The increased volatility in this scenario highlights the importance of considering supply-side factors alongside financial conditions when modeling housing markets.
- While interest rates clearly play a crucial role in housing market dynamics, this scenario demonstrates that physical market constraints can be equally, if not more, influential in determining household moving behavior.

These findings underscore the complexity of housing market dynamics and the need for models that incorporate both financial and physical market constraints to provide a comprehensive understanding of household behavior and market trends.

## 5.6 Comparison Across Scenarios

Comparing all three scenarios provides a comprehensive view of how different market conditions affect housing dynamics:

- **Market Activity:** The 3% scenario shows the highest activity, followed by the constrained scenario, and then the 6% scenario. This suggests that while low interest rates stimulate market activity, supply constraints can moderate this effect.
- **Volatility:** The constrained scenario exhibits the highest volatility, highlighting how supply-side factors can introduce instability into the market.
- **Homeownership Rates:** Counterintuitively, the constrained scenario shows the highest homeownership rate (70.6%), significantly higher than both the 3% (59.7%) and 6% (59.9%) scenarios. This suggests that supply constraints might have a more profound impact on homeownership than interest rates alone.
- **Long-term Trends:** While both unconstrained scenarios show steady increases in market activity, the constrained scenario indicates a potential saturation point, suggesting different long-term market dynamics under supply limitations.

## 5.7 Conclusion

This agent-based model simulations of the Irish housing market provide valuable insights into the complex dynamics of household moving decisions and homeownership under various conditions. The three scenarios we explored – 3% interest rate, 6% interest rate, and limited construction with restricted vacancy – reveal the multifaceted nature of housing market behavior.

Key conclusions include:

1. **Interest Rate Impact:** Lower interest rates stimulate significantly more market activity, demonstrating the powerful influence of monetary policy on housing market liquidity.
2. **Supply Constraints:** Physical market constraints can be equally, if not more, influential than interest rates in shaping market dynamics and household behavior.
3. **Homeownership Resilience:** Homeownership rates show remarkable stability across scenarios, with the constrained scenario unexpectedly showing the highest rates. This indicates that factors beyond interest rates and market activity play crucial roles in long-term housing tenure outcomes.
4. **Market Complexity:** The interplay between financial conditions (interest rates) and physical market constraints (construction and vacancy) demonstrates the need for holistic approaches in understanding and managing housing markets.

These findings have several implications for policy and further research:

- Policymakers should consider both monetary policy and supply-side interventions when aiming to influence housing market outcomes.
- The resilience and even increase of homeownership rates under supply constraints suggests that policies aimed at increasing homeownership might need to focus on factors beyond just financial accessibility or market activity.
- Future research could explore the long-term implications of sustained supply constraints on overall economic health, social mobility, and housing inequality.
- The development of more sophisticated models that can capture the complex interactions between interest rates, supply constraints, and other socioeconomic factors would be valuable for predicting and managing housing market dynamics.

In conclusion, this agent-based model provides a powerful tool for simulating and understanding the complex dynamics of the Irish housing market. By capturing the interactions between individual household decisions, broader market conditions, and structural constraints, it offers valuable insights for both policymakers and researchers. If this model is refined and expanded further, it has the potential to become an even more robust platform for testing housing market interventions and predicting long-term market trends.

## 5.8 Discussion

This discussion reflects on the agent-based model of the Irish housing market, highlighting its theoretical alignment with complex adaptive systems theory and its methodological advantages in capturing micro-level behaviors. It acknowledges ethical considerations in using such models for policy decisions and suggests future research directions, including spatial dynamics integration and policy scenario testing. The section emphasizes the model’s contributions to housing market understanding while recognizing areas for further development and refinement.

### 5.8.1 Theoretical Implications

My agent-based model (ABM) of the Irish housing market provides several insights that contribute to the broader understanding of housing markets. The emergent patterns observed in my simulation, particularly the dynamics of housing tenure shifts and price fluctuations, align with the complex adaptive systems theory as applied to urban housing markets [25]. This theoretical framework suggests that housing markets are characterized by non-linear interactions between heterogeneous agents, leading to emergent phenomena that cannot be predicted by analyzing individual components in isolation.

My findings support the notion that housing markets are inherently dynamic and subject to feedback loops. For instance, the observed relationship between housing affordability and household mobility aligns with the "housing career" concept proposed by Pickles and Davies [26], which posits that households make strategic housing choices based on their changing needs and market conditions over time.

Furthermore, the inclusion of a population model in my simulation, absent in comparable works such as Kerkhoven’s [23], allows for a more comprehensive exploration of demographic impacts on housing market dynamics. This addition supports the life-cycle theory of housing demand [27], demonstrating how demographic shifts can significantly influence housing market outcomes.

### 5.8.2 Methodological Reflections

The use of agent-based modeling in this study offers several advantages over traditional econometric approaches to housing market analysis. ABM allows for a bottom-up approach that captures the heterogeneity of households and the complexity of their decision-making processes, which is challenging to incorporate in equation-based models [5].

My model’s ability to simulate individual household decisions and their aggregate effects on the market provides insights into micro-level behaviors that drive macro-level outcomes. This aligns with the strengths of ABM identified by Crooks et al. [28] in geographical applications, particularly in capturing spatial and temporal dynamics of complex systems.

However, it’s important to acknowledge the limitations of this approach. The validity of ABM results heavily depends on the accuracy of the underlying assumptions and rules governing agent behavior. As noted by O’Sullivan et al. [29], there’s often a trade-off between model complexity and interpretability. My model strikes a balance, incorporating key elements like a population model while maintaining tractability.

Compared to Kerkhoven’s model [?], my inclusion of a population model represents a significant methodological advancement. This addition allows for a more realistic representation of demographic changes and their impacts on housing demand and supply, addressing a key limitation in previous housing market ABMs.

### 5.8.3 Ethical Considerations

The use of agent-based models in housing market research raises several ethical considerations. Firstly, there’s the potential for model outcomes to influence policy decisions that have real-world impacts on housing accessibility and affordability. As emphasized by Dawid and Gatti [30], researchers have a responsibility to clearly communicate the limitations and assumptions of their models to prevent misinterpretation or misuse of results.

Additionally, my model’s treatment of social housing and its impact on low-income households highlights the ethical dimensions of housing policy.

The simulation results showing patterns of segregation or concentrations of disadvantage echo concerns raised by Galster [31] about the unintended consequences of housing policies. This underscores the need for careful consideration of equity and social justice in both model design and the interpretation of results.

Furthermore, the use of individual-level data in ABMs, even when anonymized, raises privacy concerns. While my model uses synthetic data in some parts, the increasing availability of big data in urban research necessitates ongoing ethical scrutiny, as discussed by Kitchen [32].

#### **5.8.4 Future Research Directions**

This study opens several avenues for future research:

1. **Integration of Spatial Dynamics:** Incorporating geographical information system (GIS) data to model spatial aspects of the housing market could provide insights into neighborhood effects and locational preferences [33].
2. **Enhanced Behavioral Models:** Developing more sophisticated decision-making algorithms for agents, possibly incorporating machine learning techniques, could improve the realism of household behaviors [34].
3. **Policy Scenario Testing:** Expanding the model to simulate various policy interventions (e.g., rent controls, zoning changes) could provide valuable insights for policymakers [35].
4. **Long-term Demographic Projections:** Extending the population model to incorporate long-term demographic trends could help forecast housing needs over extended periods [36].
5. **Environmental Factors:** Integrating environmental considerations, such as energy efficiency of housing stock or climate change impacts, could address the growing intersection of housing and sustainability issues [37].

These future directions would not only enhance the model's predictive capabilities but also broaden its applicability to a wider range of housing market challenges.

# Appendix A

## Code

This appendix provides the code for key functions from each main class in the Irish Housing ABM simulation. These snippets represent the core functionality of the model and are crucial for understanding the implementation details.

### A.0.1 Simulation Class

#### A.0.1.1 Run Method

This method is the main driver of the simulation, executing the model for a specified number of years.

```
1 public void Run(int years, double scaleFactor)
2 {
3     Initialize(scaleFactor);
4     InitializeExcelHeader();
5
6     for (int year = 0; year < years; year++)
7     {
8         Console.WriteLine($"Starting year {year + 1}");
9         householdMoves = 0; // Reset moves counter at the
            start of each year
10
11         for (int month = 0; month < 12; month++)
12         {
```

```

13         Console.WriteLine($" Processing month {month +
14             1}");
15         simulationUpdater.PerformMonthlyUpdates(households,
16             houses, ref householdMoves);
17         OutputMonthlyStatistics(year, month);
18         Console.WriteLine($" Completed month {month +
19             1} of year {year + 1}");
20     }
21
22     double taxRate = 0.2; // Example tax rate
23     simulationUpdater.PerformYearlyUpdates(households,
24         houses, ref taxRate);
25
26     OutputYearlyStatistics(year, 11); // Output yearly
27         statistics after December (month 11)
28
29     Console.WriteLine($"Completed year {year + 1}");
30 }
31
32 // Save and close Excel file
33 worksheet.Cells[worksheet.Dimension.Address].AutoFitColumns();
34 var yearlySheet =
35     excelPackage.Workbook.Worksheets["Yearly Statistics"];
36 if (yearlySheet != null)
37 {
38     yearlySheet.Cells[yearlySheet.Dimension.Address].AutoFitColumns();
39 }
40 excelPackage.Save();
41 Console.WriteLine("Simulation complete");
42 }

```

## A.0.2 SimulationUpdater Class

### A.0.2.1 PerformMonthlyUpdates Method

This method handles all the monthly update processes in the simulation.

```

1 public void PerformMonthlyUpdates(List<Household>
2     households, List<House> houses, ref int householdMoves)
3 {

```

```

3      Console.WriteLine("    Updating house prices");
4      UpdateHousePrices(houses);
5      Console.WriteLine("    Updating contracts");
6      UpdateContracts(households);
7      Console.WriteLine("    Deciding to move");
8      DecideToMove(households, houses);
9      Console.WriteLine("    Acting on best options");
10     ActOnBestOptions(households, houses, ref householdMoves);
11     Console.WriteLine("    Assigning social housing");
12     AssignSocialHousing(households, houses, ref
        householdMoves);
13     Console.WriteLine("    Processing new listings");
14     ProcessNewListings(houses);
15     Console.WriteLine("    Collecting bank payments");
16     bank.CollectPayments();
17 }

```

### A.0.2.2 DecideToMove Method

This method implements the decision-making process for households considering a move.

```

1 private void DecideToMove(List<Household> households,
    List<House> houses)
2 {
3     List<House> availableHouses = houses.Where(h =>
        h.IsAvailable()).ToList();
4     Console.WriteLine($"    Available Houses:
        {availableHouses.Count}");
5
6     double baseMovementCost = 2.0;
7
8     int householdsConsidered = 0;
9     int householdsWantingToMove = 0;
10
11     foreach (var household in households)
12     {
13         household.DecideToMove(availableHouses,
            baseMovementCost, CanAffordHouse,
            bank.CalculateMonthlyPayment);

```

```

14
15     householdsConsidered++;
16     if (household.WantToMove) householdsWantingToMove++;
17
18     if (householdsConsidered % 1000 == 0 ||
        householdsConsidered == households.Count)
19     {
20         Console.WriteLine($"          Processed
            {householdsConsidered}/{households.Count}
            households. {householdsWantingToMove} want to
            move.");
21     }
22 }
23 Console.WriteLine("      Completed DecideToMove");
24 }

```

## A.0.3 Household Class

### A.0.3.1 DecideToMove Method

This method determines whether a household wants to move based on utility calculations.

```

1 public void DecideToMove(List<House> suitableHouses, double
    baseMovementCost, Func<Household, House, bool>
    canAffordHouseFunc, Func<double, int, double, double>
    calculateMonthlyPaymentFunc)
2 {
3     House currentHouse = this.Contract?.House;
4     double currentUtility = CalculateUtility(currentHouse,
        0);
5
6     double bestUtility = currentUtility;
7     House bestHouse = null;
8
9     foreach (var house in suitableHouses)
10    {
11        if (canAffordHouseFunc(this, house))
12        {

```

```

13         double movementCost =
                CalculateMovementCost(baseMovementCost,
                currentHouse, house);
14         double utility = CalculateUtility(house,
                movementCost);
15
16         if (utility > bestUtility)
17         {
18             bestUtility = utility;
19             bestHouse = house;
20         }
21     }
22 }
23
24 // Calculate the percentage improvement in utility
25 double percentageImprovement = (bestUtility -
    currentUtility) / currentUtility;
26
27 // Calculate a probability of moving based on the
    percentage improvement
28 double moveProbability = Math.Min(1,
    percentageImprovement * 3); // Adjust the multiplier
    as needed
29
30 // Decide to move based on the calculated probability
31 WantToMove = RandomNumberGenerator.NextDouble() <
    moveProbability;
32 }

```

## A.0.4 House Class

### A.0.4.1 UpdatePrice Method

This method updates the price of a house based on market conditions.

```

1 public void UpdatePrice(double factor)
2 {
3     Price *= factor;
4 }

```

## A.0.5 Bank Class

### A.0.5.1 ApproveMortgage Method

This method determines whether a mortgage is approved for a household.

```
1 public bool ApproveMortgage(Household household, double
   housePrice, double downPayment, int loanTermYears)
2 {
3     double loanAmount = housePrice - downPayment;
4     double loanToValueRatio = loanAmount / housePrice;
5     double monthlyPayment =
        CalculateMonthlyPayment(loanAmount, loanTermYears,
        InterestRate);
6
7     if (loanToValueRatio > MaxLoanToValueRatio)
8         return false;
9
10    if (household.TotalIncome < MaxDebtToIncomeRatio *
        monthlyPayment * 12)
11        return false;
12
13    household.LoanAmount = loanAmount;
14    household.LoanTermYears = loanTermYears;
15    MortgagedHouseholds.Add(household);
16    return true;
17 }
```

## A.0.6 ConstructionCompany Class

### A.0.6.1 ConstructNewHouses Method

This method handles the construction of new houses based on market conditions.

```
1 public void ConstructNewHouses(List<House> houses,
   List<Household> households)
2 {
3     int totalHouses = houses.Count;
4     int vacantHouses = houses.Count(h => h.IsAvailable());
```

```

5      double currentVacancyRate = (double)vacantHouses /
        totalHouses;
6
7      const double TargetVacancyRate = 0.05; // 5% vacancy rate
8
9      int baseHouseCount = (int)(BaseAnnualNewHouseCount /
        scaleFactor);
10
11     double vacancyAdjustment = (TargetVacancyRate -
        currentVacancyRate) * totalHouses;
12     int adjustedHouseCount = baseHouseCount +
        (int)vacancyAdjustment;
13
14     double randomFactor = 0.8 +
        (RandomNumberGenerator.NextDouble() * 0.4); // Random
        factor between 0.8 and 1.2
15     int finalHouseCount = Math.Max(0,
        (int)(adjustedHouseCount * randomFactor));
16
17     Console.WriteLine($"Constructing {finalHouseCount} new
        houses. Current vacancy rate:
        {currentVacancyRate:P2}");
18
19     for (int i = 0; i < finalHouseCount; i++)
20     {
21         double size = DrawSizeFromDistribution();
22         int quality = DrawQualityFromDistribution();
23         HouseType type = DetermineHouseType();
24         double initialPrice = CalculateInitialPrice(size,
            quality, type);
25         var newHouse = new House(size, quality, type,
            initialPrice);
26         if (HouseIsSuitableForMarket(newHouse, households))
27         {
28             households.Add(newHouse);
29             OnHouseConstructed(newHouse);
30         }
31     }
32 }

```

These code snippets provide a detailed look at the core functionalities of the Irish Housing ABM simulation. They showcase the implementation of key processes such as running the simulation, updating the model state, decision-making for households, mortgage approval, and new house construction. This appendix serves as a valuable reference for understanding the technical details of the model's implementation.

The full code for this project can be found here: <https://github.com/GingerBrains/irish-housing-abm>

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